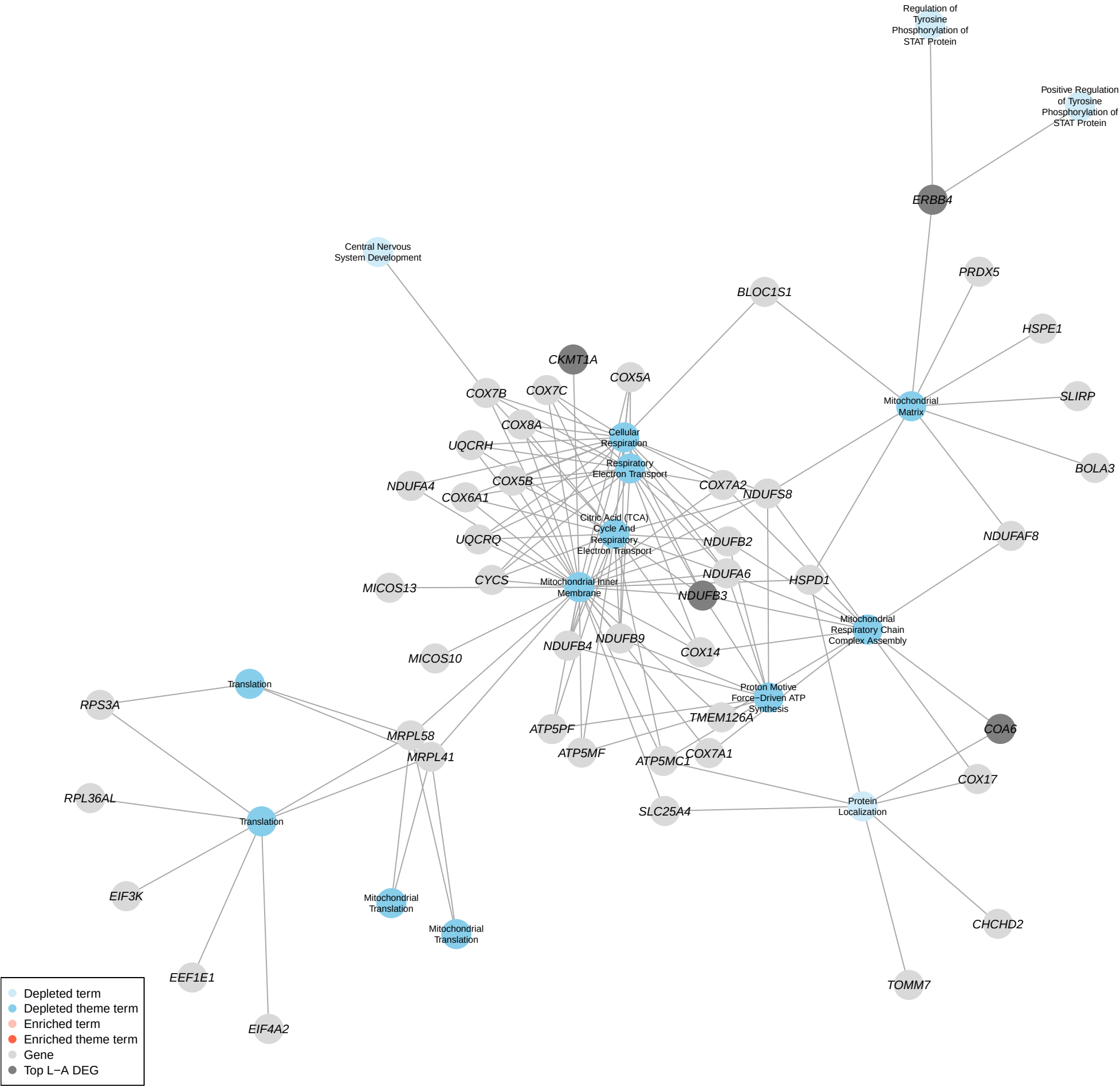
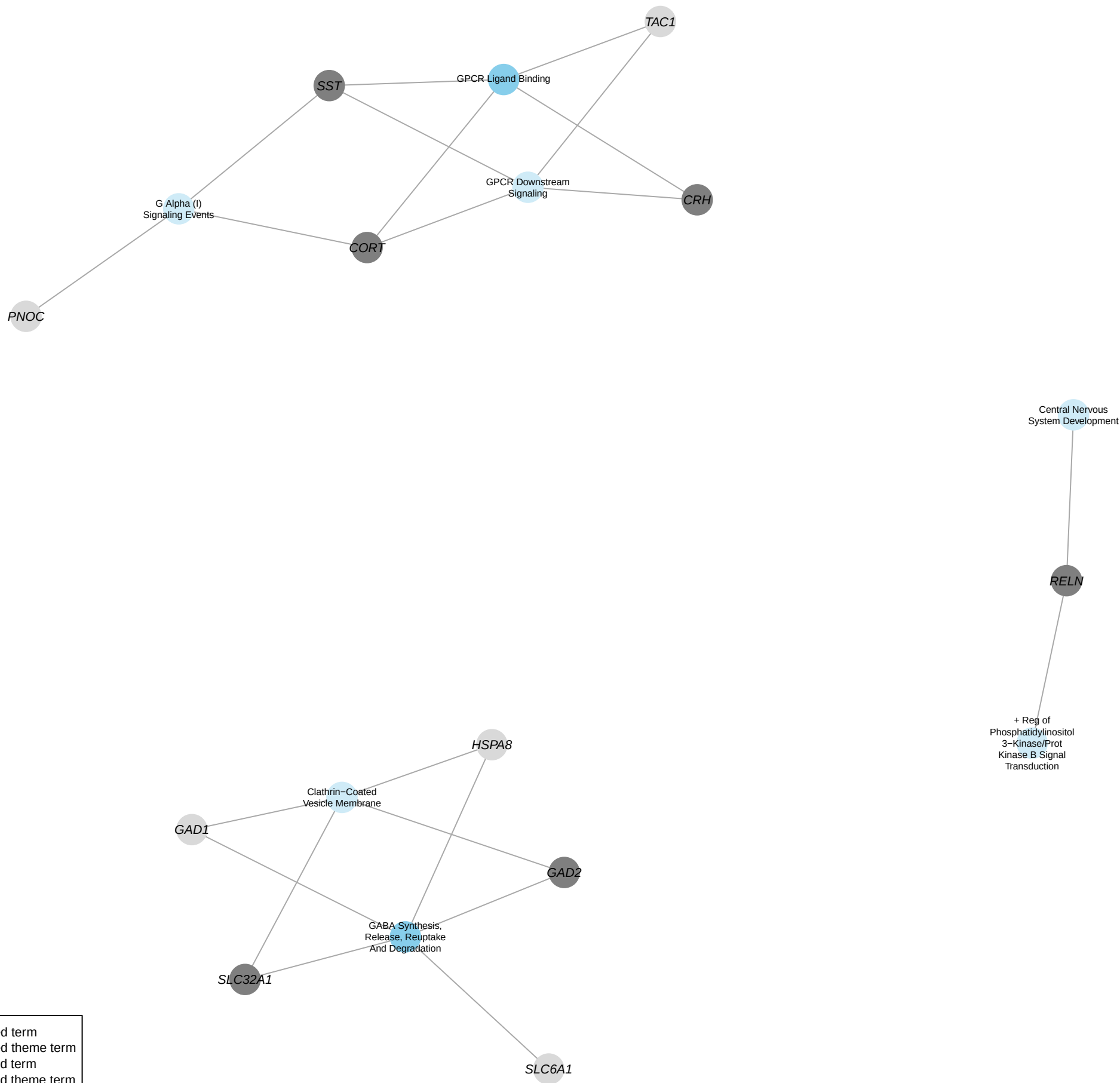


NTC.BPD M upkeep theme

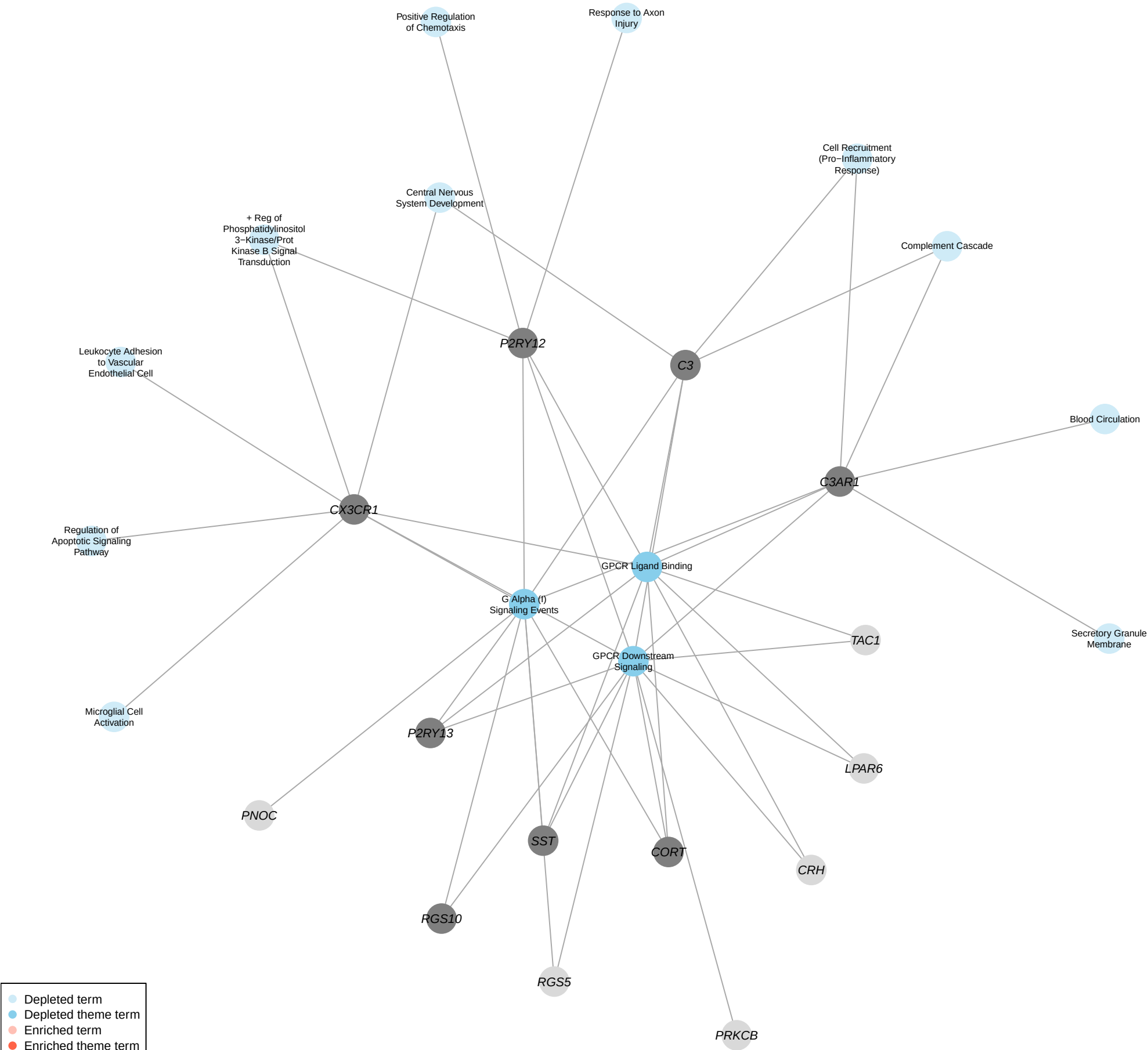


NTC.BPD M gaba theme

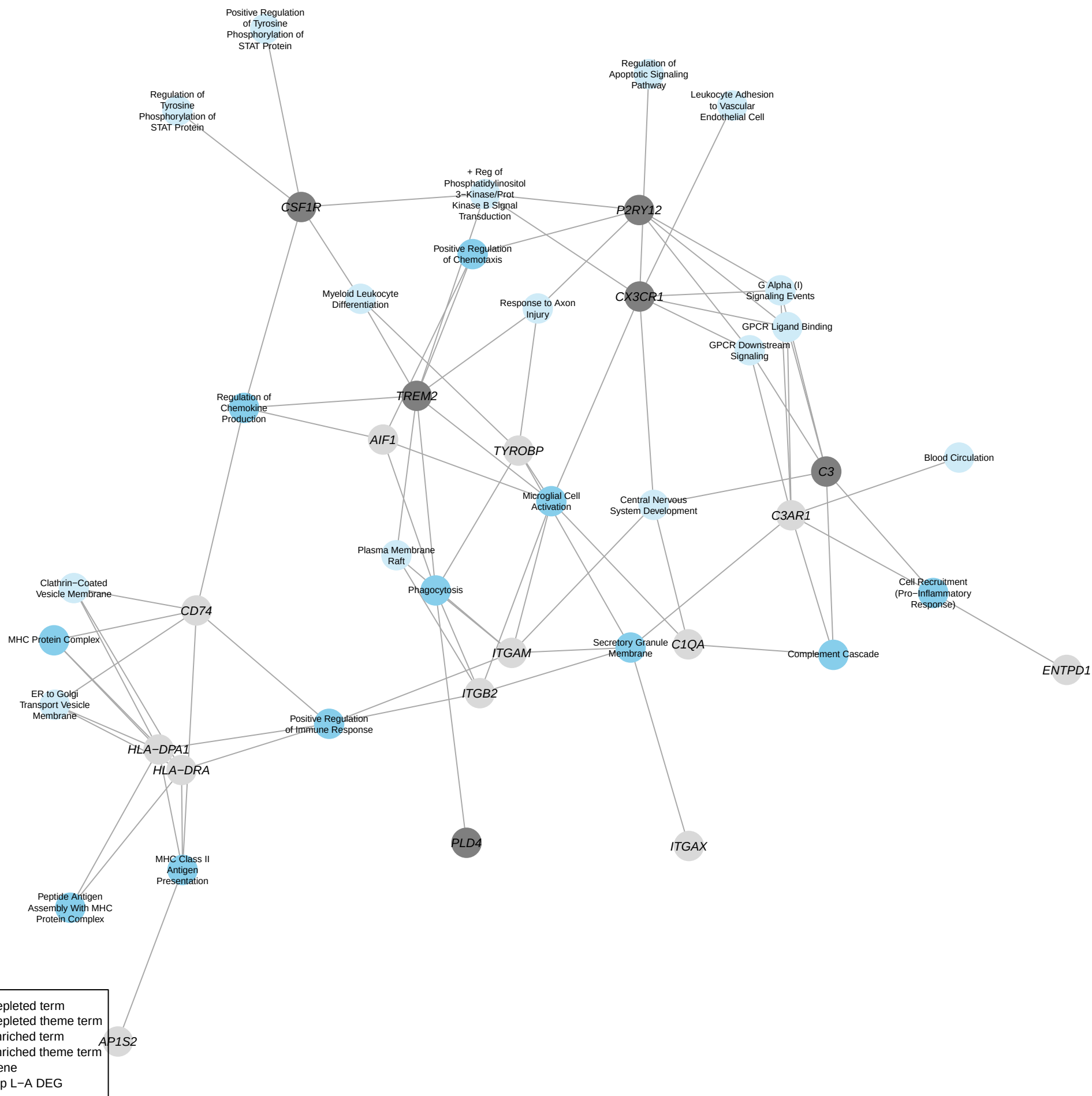


- Depleted term
- Depleted theme term
- Enriched term
- Enriched theme term
- Gene
- Top L-A DEG

NTC.BPD M gpcr theme

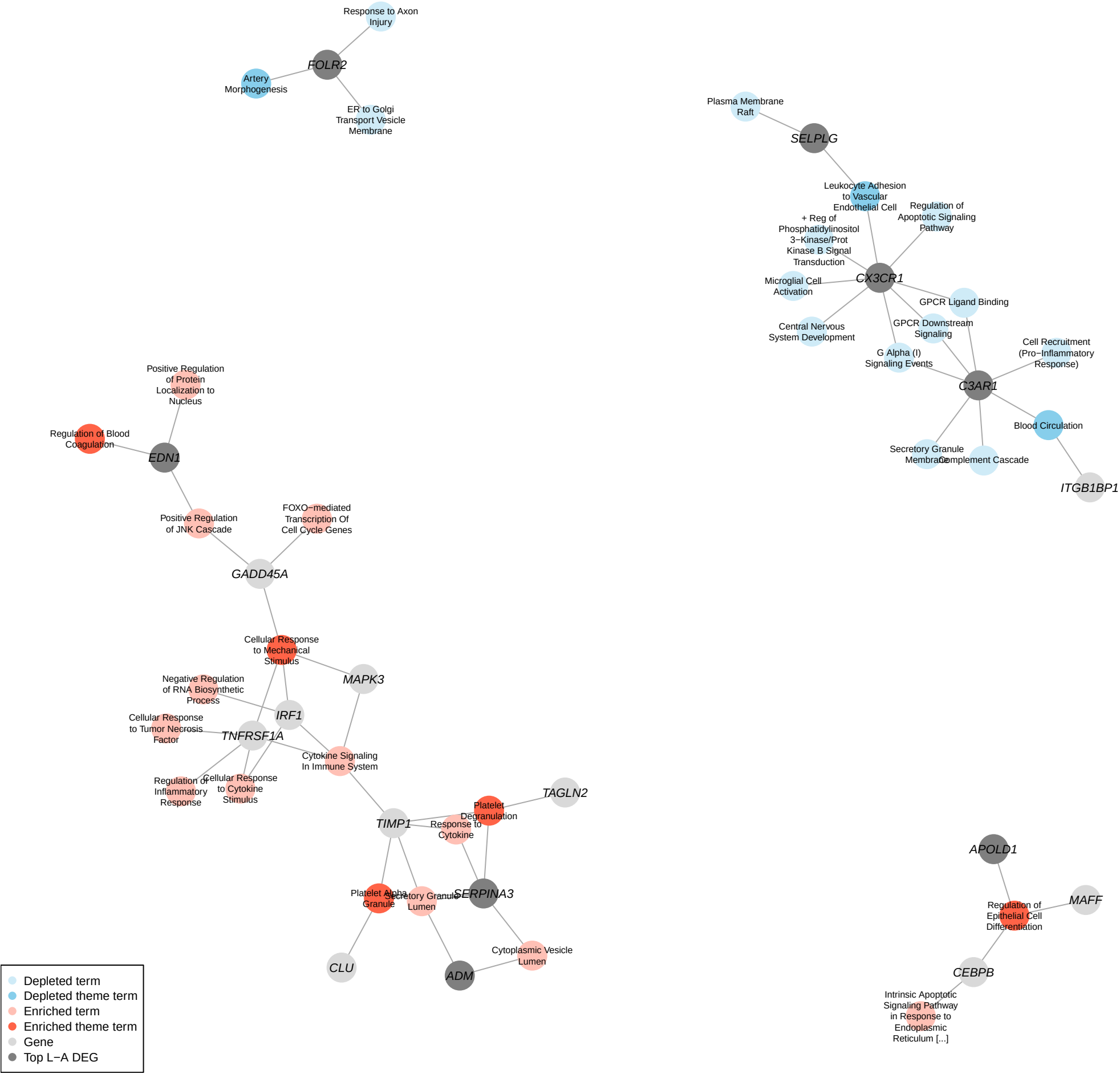


NTC.BPD M immune theme



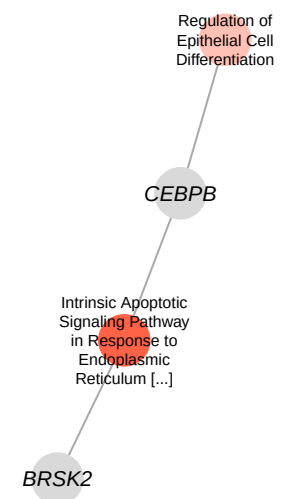
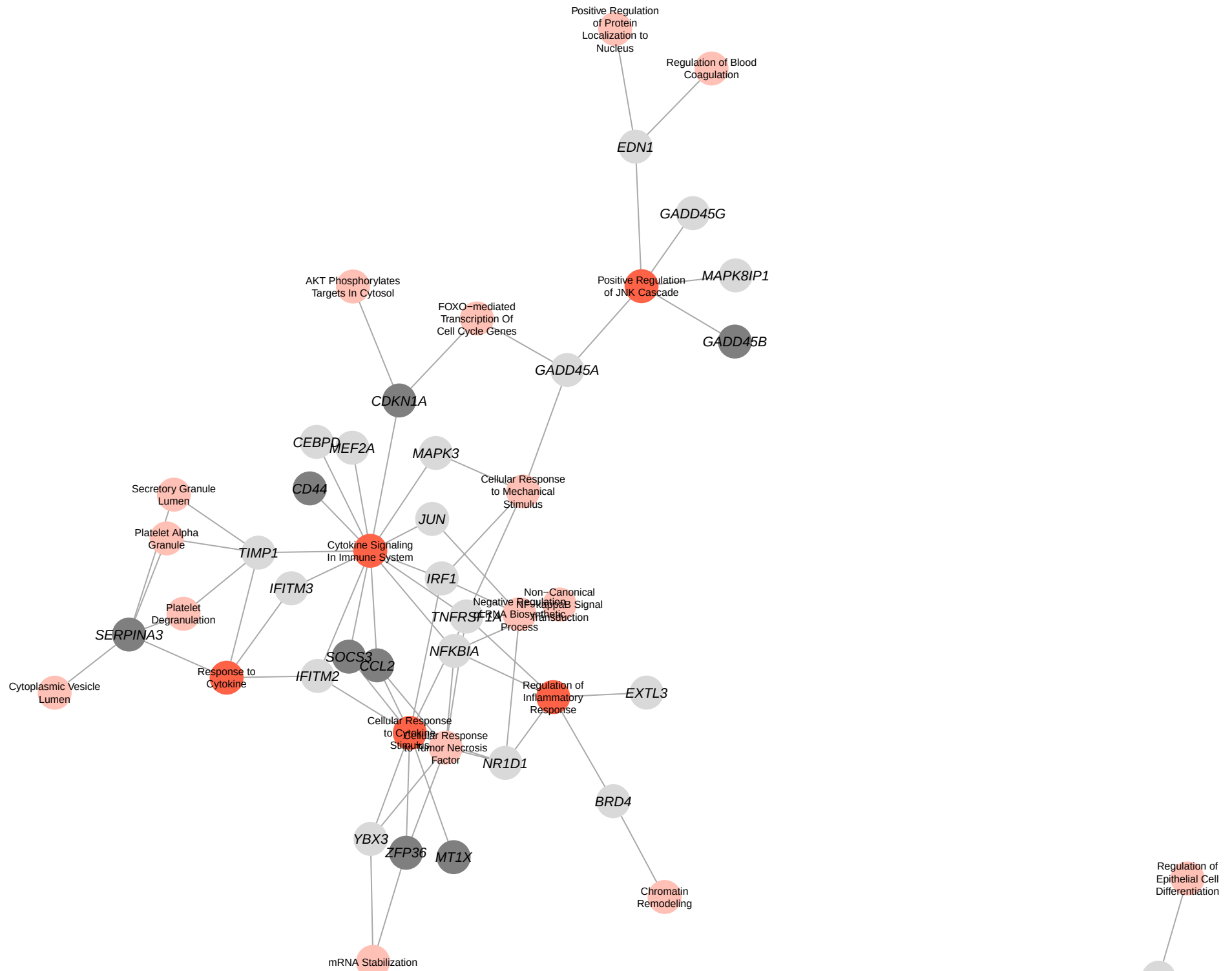
Reactome, GO-BP, and GO-CC results combined to produce theme

NTC.BPD M vascular theme



Reactome, GO-BP, and GO-CC results combined to produce theme

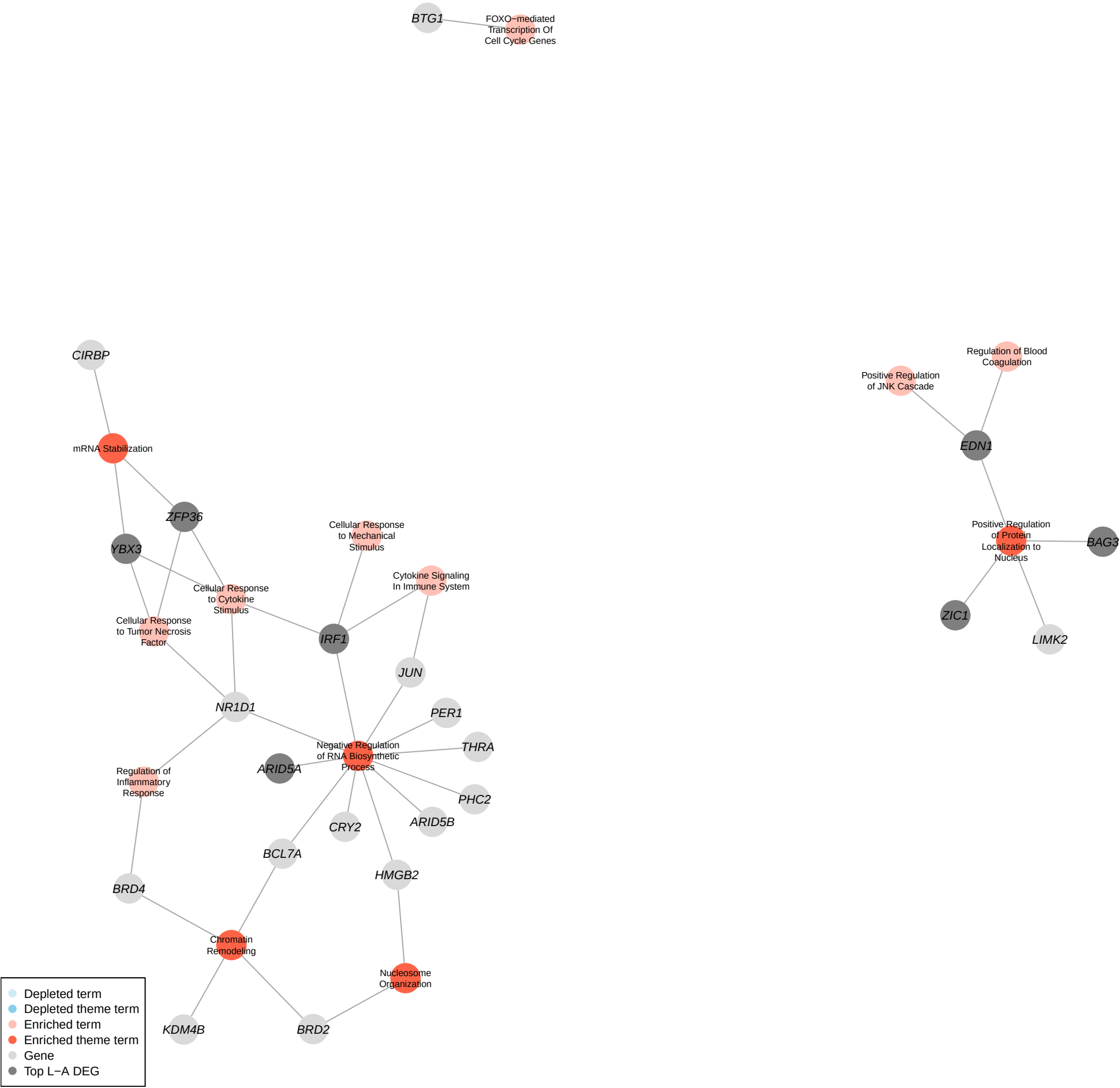
NTC.BPD M proinfl theme



- Depleted term
- Depleted theme term
- Enriched term
- Enriched theme term
- Gene
- Top L-A DEG

Reactome, GO-BP, and GO-CC results combined to produce theme

NTC.BPD M nuclear theme



Reactome, GO-BP, and GO-CC results combined to produce theme

Intersection Size

50
40
30
20
10
0

46

15

15

14

8

6

5

6

5

5

2

2

2



gaba



gpcr



vascular



immune



nuclear



proinfl



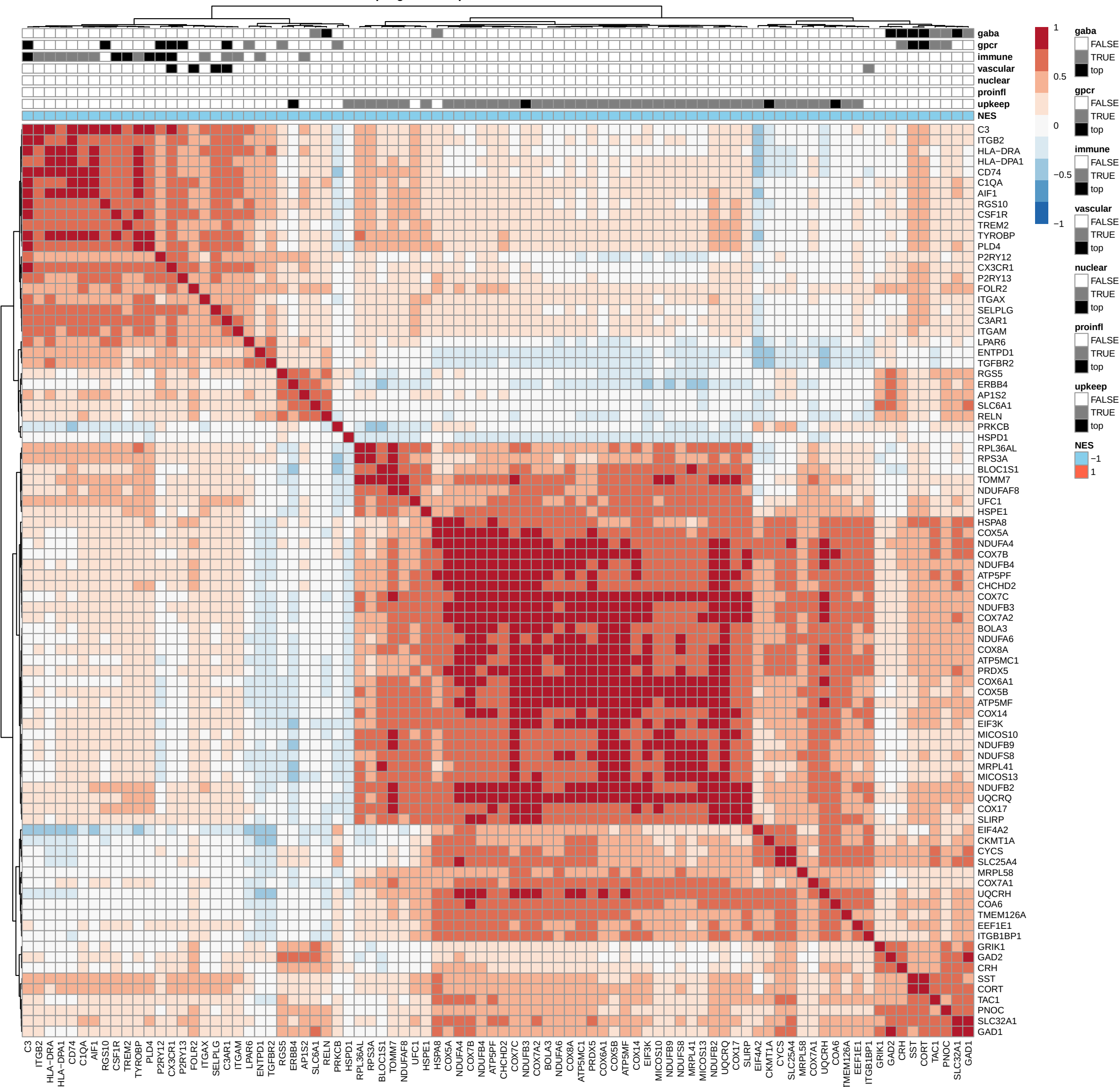
upkeep



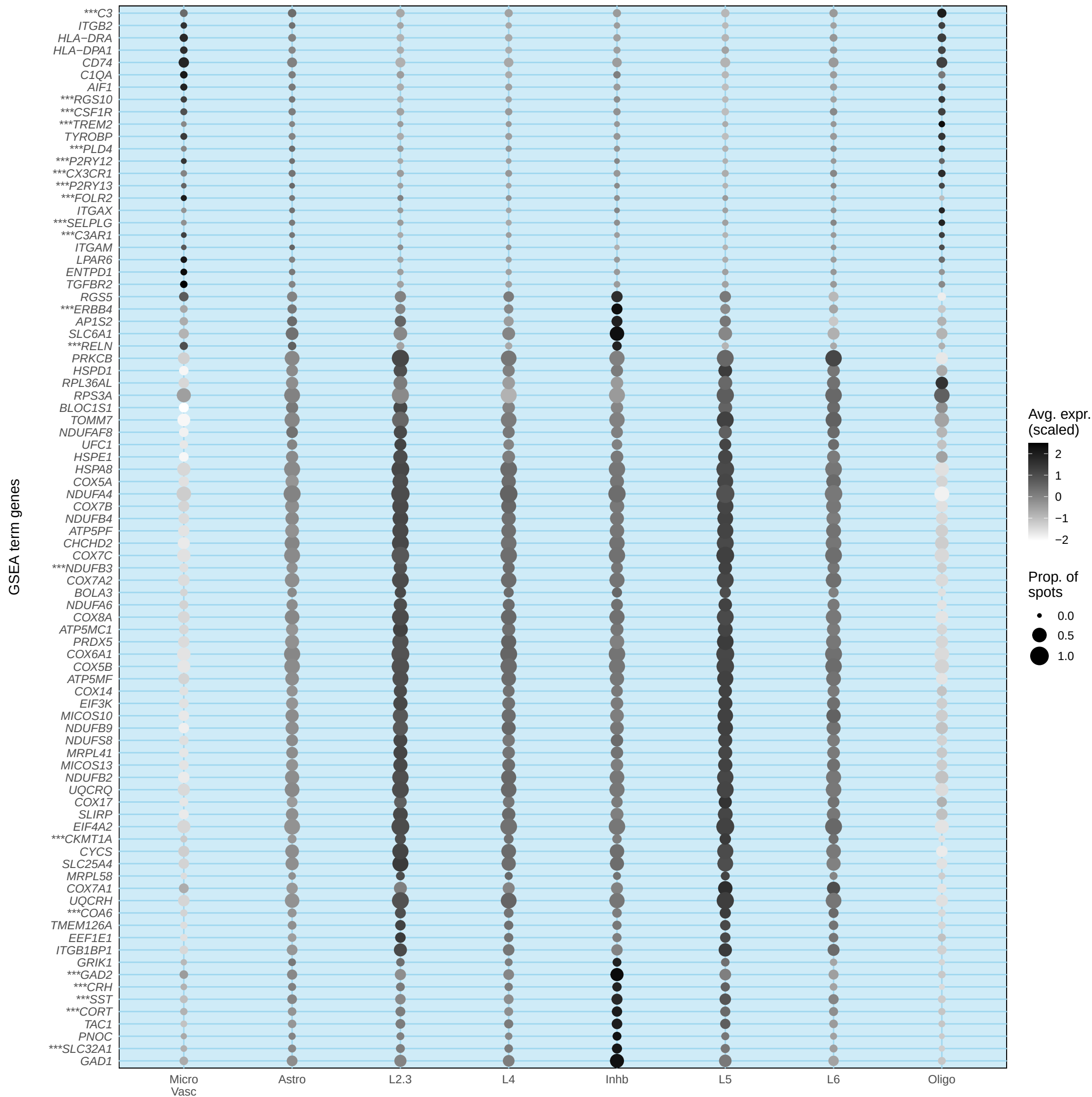
Set Size

40 30 20 10 0

NTC.BPD M L-A depleted genes
Top logFC DEGs per theme annotated with black



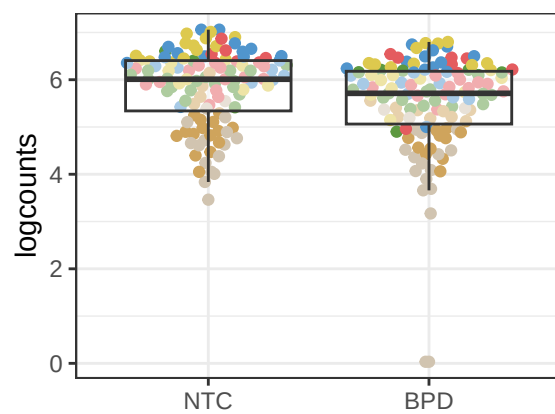
NTC.BPD M L-A depleted terms (***)top L-A DEGs per theme)



CKMT1A

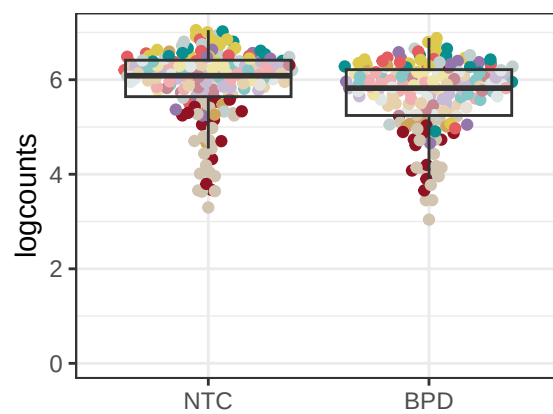
PRECAST (smoothed)

L-A sig: adj p< 5e-02 logFC = -0.27



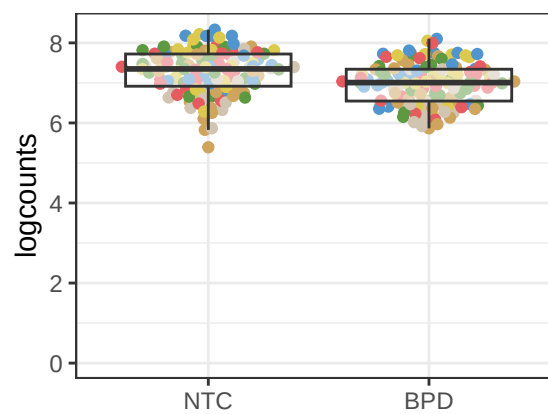
Seurat labels

L-A sig: adj p< 3e-02 logFC = -0.28

**NDUFB3**

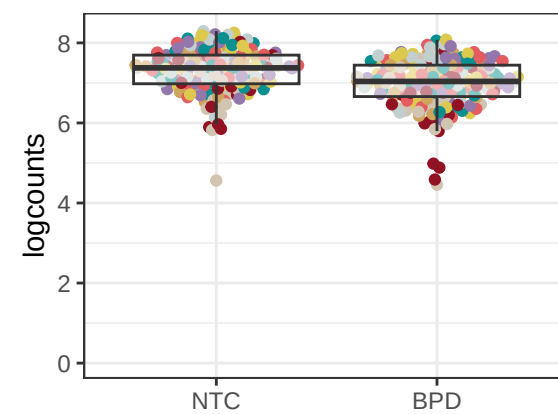
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = -0.26



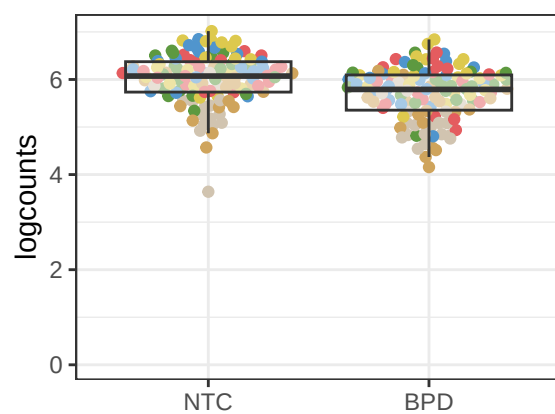
Seurat labels

L-A sig: adj p< 6e-03 logFC = -0.25

**COA6**

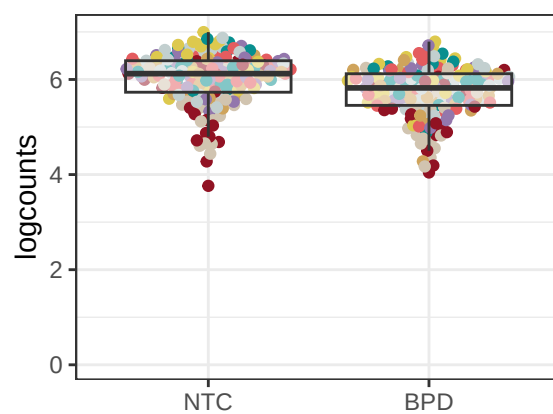
PRECAST (smoothed)

L-A sig: adj p< 2e-02 logFC = -0.25



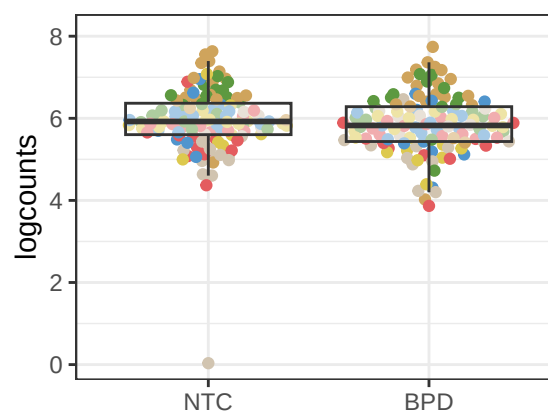
Seurat labels

L-A sig: adj p< 9e-03 logFC = -0.28

**ERBB4**

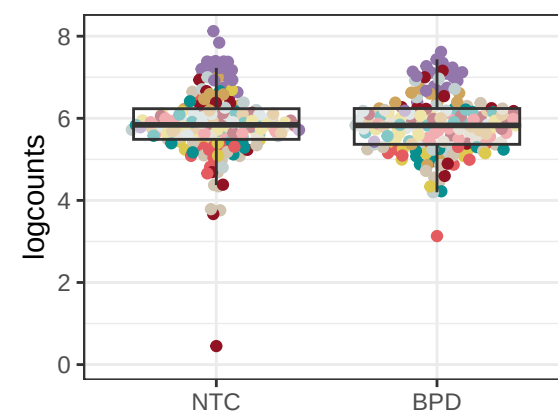
PRECAST (smoothed)

L-A sig: adj p< 5e-02 logFC = -0.23



Seurat labels

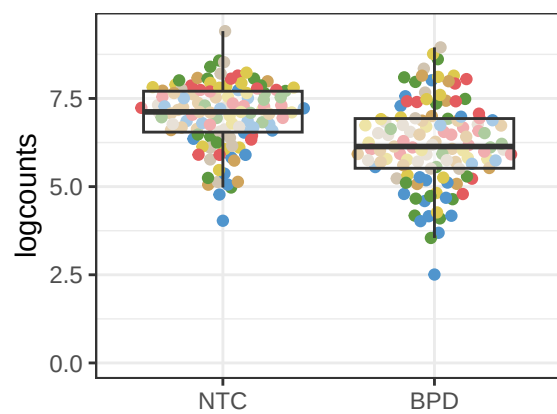
L-A sig: adj p< 2e-02 logFC = -0.25



SST

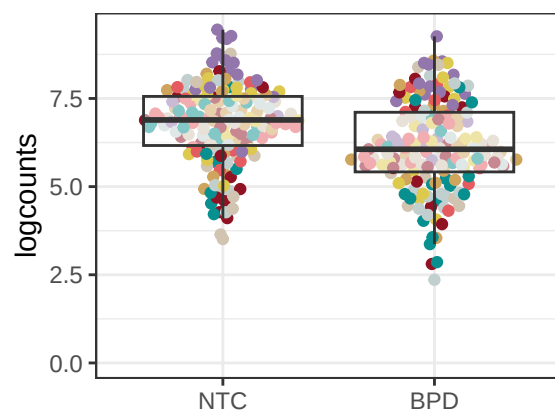
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = -0.81



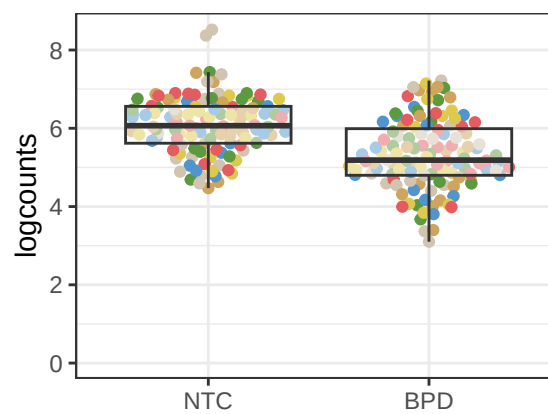
Seurat labels

L-A sig: adj p < 1e-03 logFC = -0.75

**CORT**

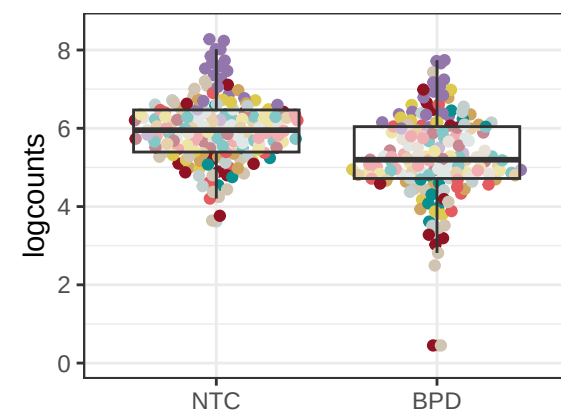
PRECAST (smoothed)

L-A sig: adj p < 2e-04 logFC = -0.61



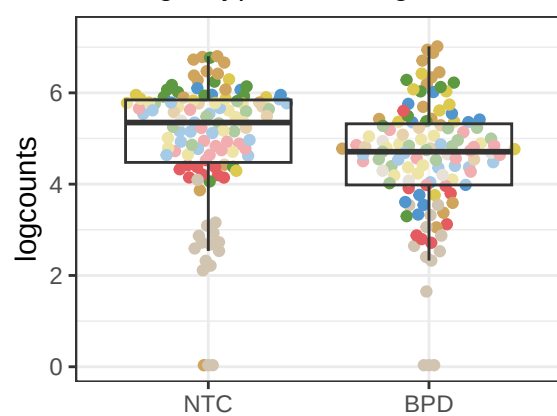
Seurat labels

L-A sig: adj p < 2e-04 logFC = -0.57

**CRH**

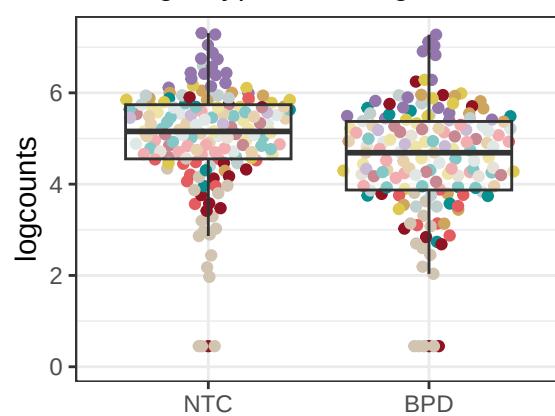
PRECAST (smoothed)

L-A sig: adj p < 2e-04 logFC = -0.61



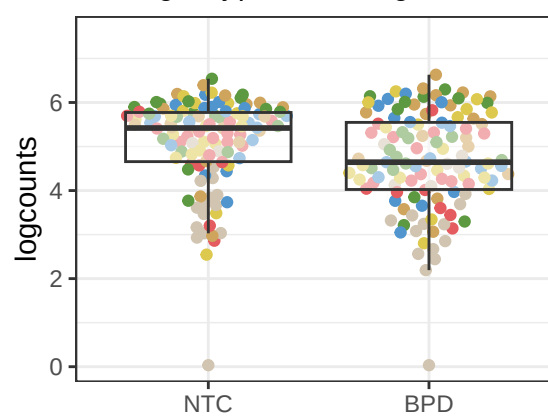
Seurat labels

L-A sig: adj p < 1e-03 logFC = -0.56

**SLC32A1**

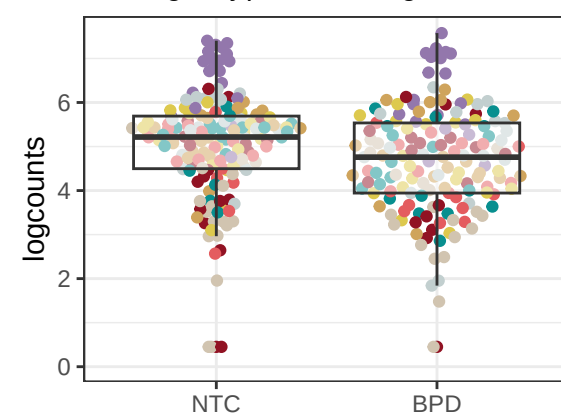
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.49



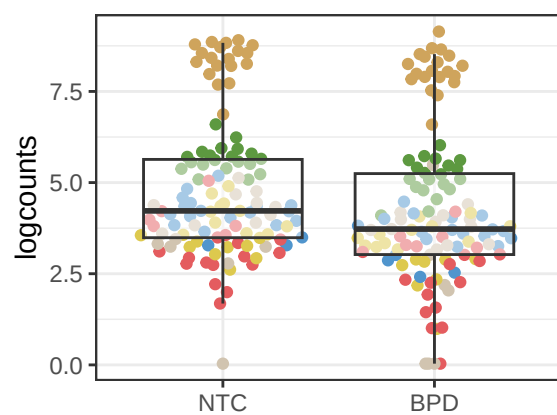
Seurat labels

L-A sig: adj p < 3e-02 logFC = -0.49

**RELN**

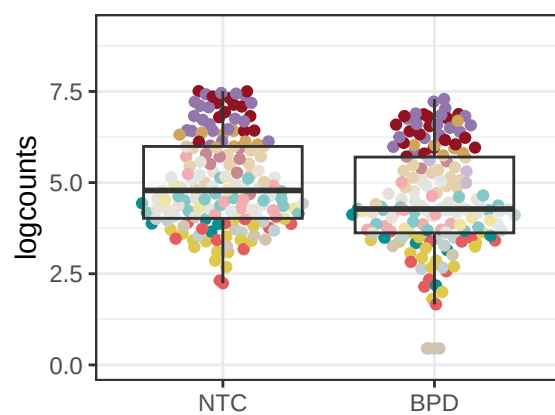
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.43



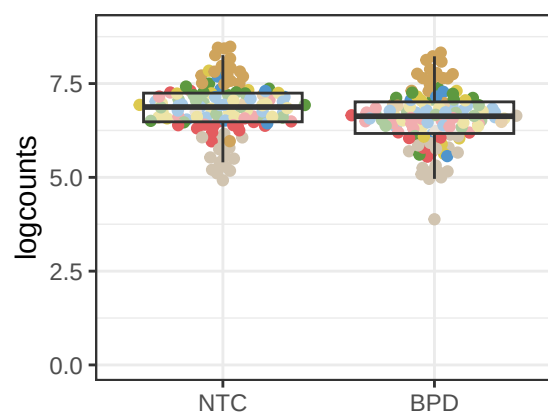
Seurat labels

L-A sig: adj p < 3e-02 logFC = -0.4

**GAD2**

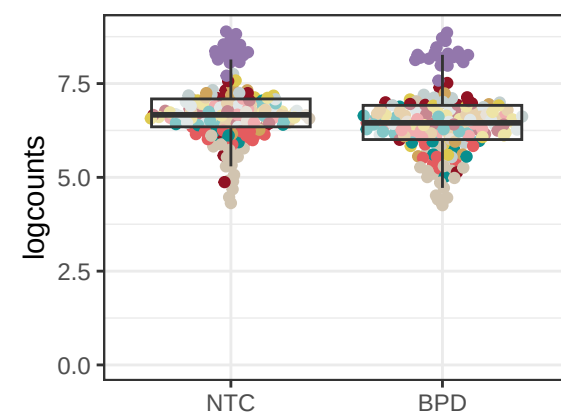
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.28



Seurat labels

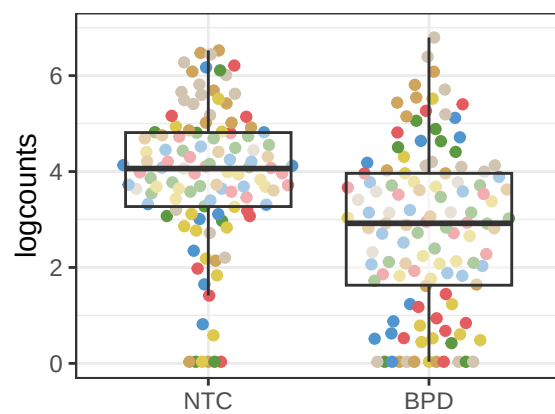
L-A sig: adj p < 1e-02 logFC = -0.29



CX3CR1

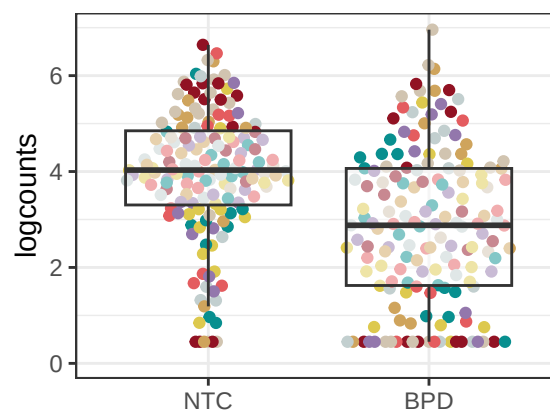
PRECAST (smoothed)

L-A sig: adj p < 1e-07 logFC = -1.52



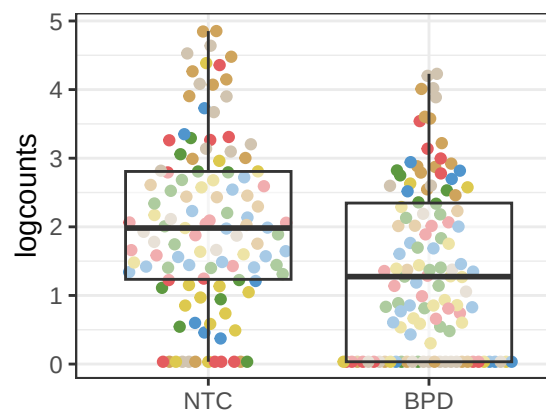
Seurat labels

L-A sig: adj p < 2e-08 logFC = -1.47

**P2RY13**

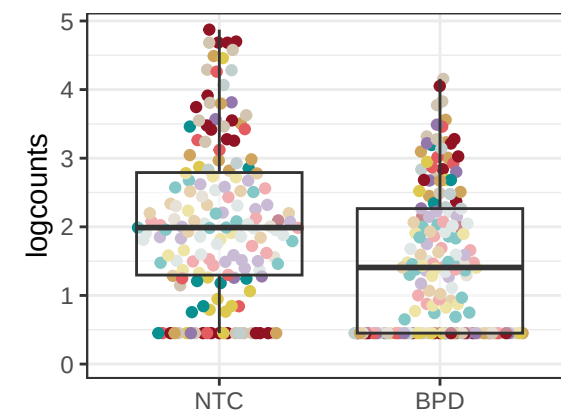
PRECAST (smoothed)

L-A sig: adj p < 4e-03 logFC = -1.03



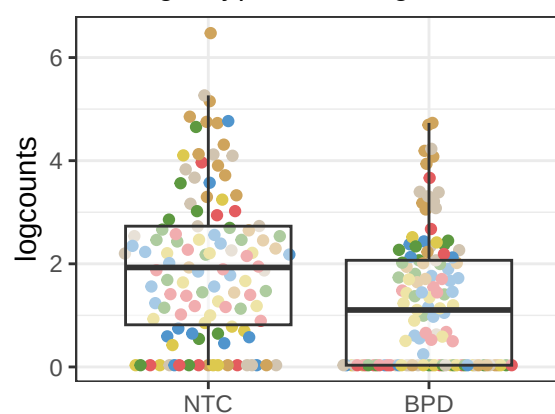
Seurat labels

L-A sig: adj p < 6e-03 logFC = -0.92

**P2RY12**

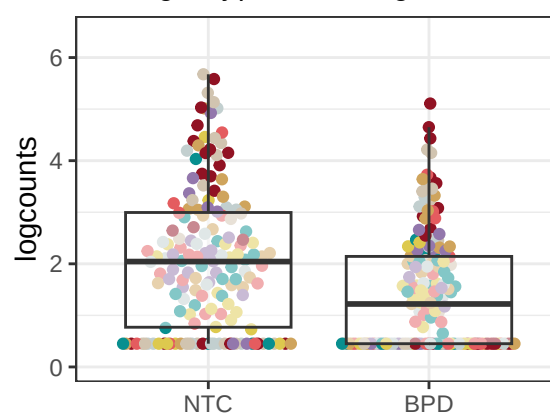
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.97



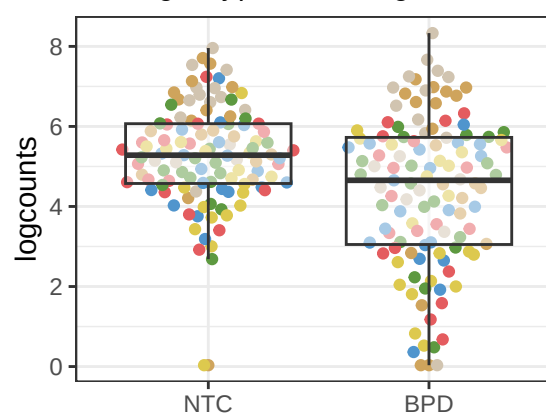
Seurat labels

L-A sig: adj p < 1e-02 logFC = -0.95

**C3**

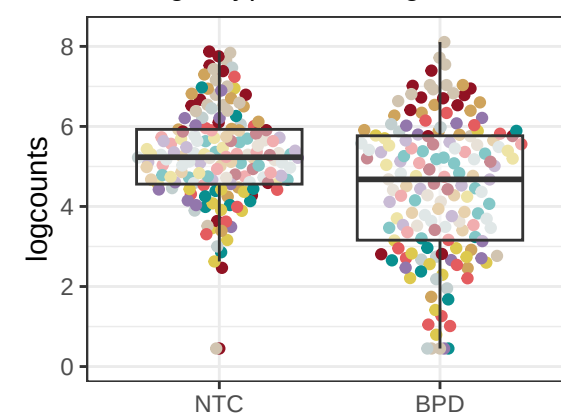
PRECAST (smoothed)

L-A sig: adj p < 3e-04 logFC = -0.97



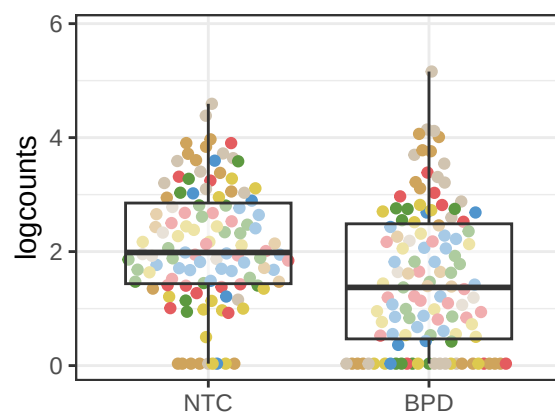
Seurat labels

L-A sig: adj p < 2e-05 logFC = -1

**C3AR1**

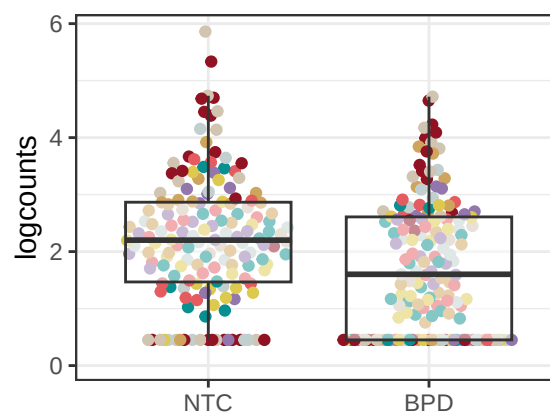
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.87



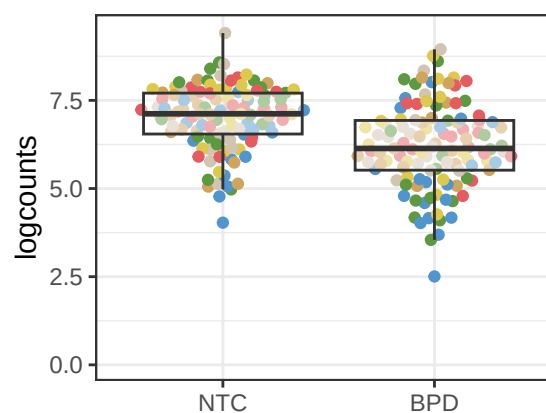
Seurat labels

L-A sig: adj p < 3e-03 logFC = -1

**SST**

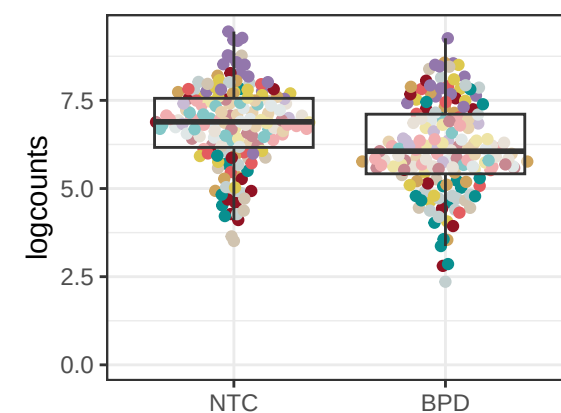
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = -0.81



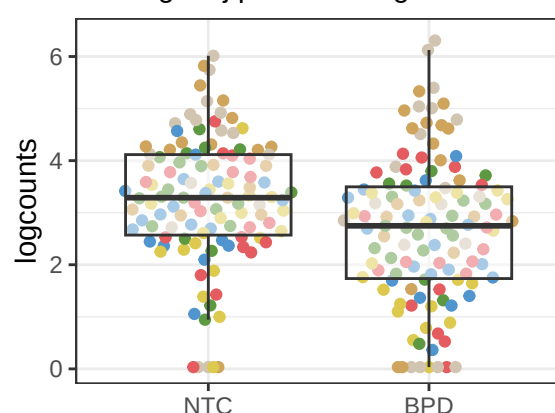
Seurat labels

L-A sig: adj p < 1e-03 logFC = -0.75

**RGS10**

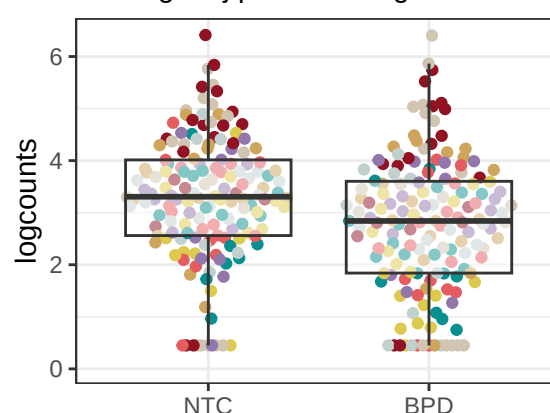
PRECAST (smoothed)

L-A sig: adj p < 4e-03 logFC = -0.71



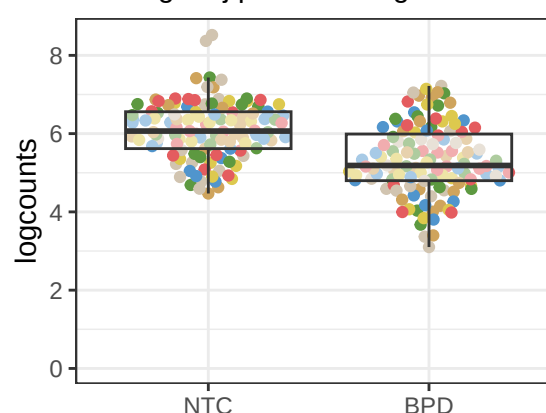
Seurat labels

L-A sig: adj p < 4e-03 logFC = -0.72

**CORT**

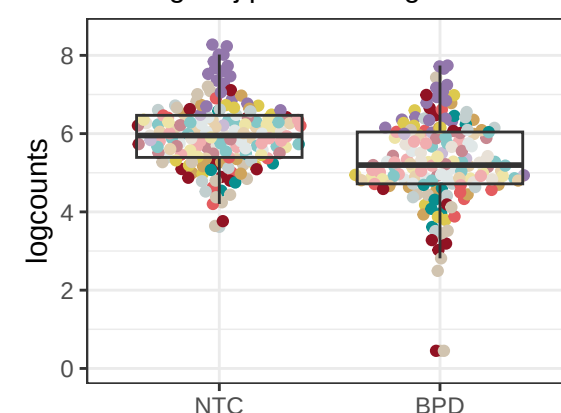
PRECAST (smoothed)

L-A sig: adj p < 2e-04 logFC = -0.61



Seurat labels

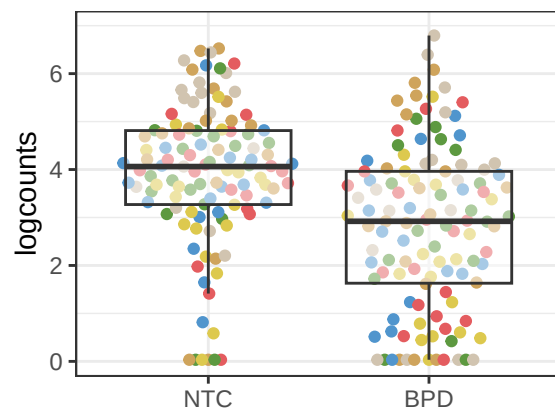
L-A sig: adj p < 2e-04 logFC = -0.57



CX3CR1

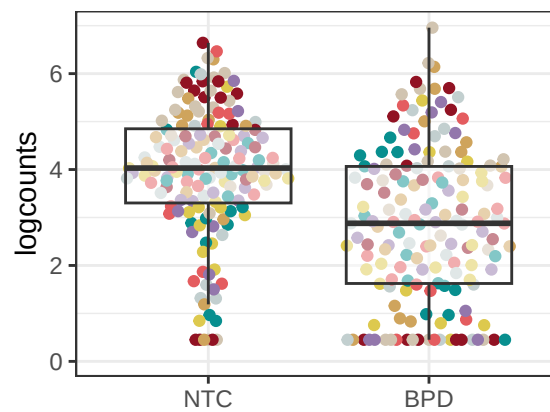
PRECAST (smoothed)

L-A sig: adj p < 1e-07 logFC = -1.52



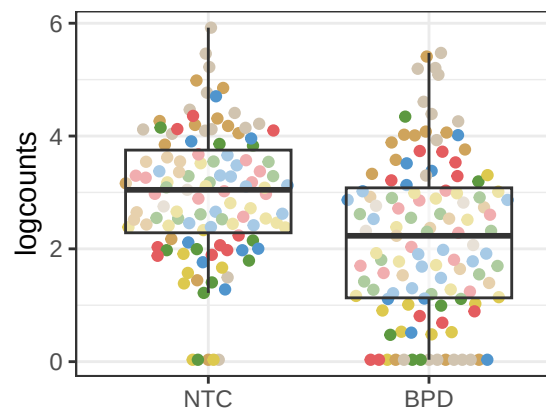
Seurat labels

L-A sig: adj p < 2e-08 logFC = -1.47

**PLD4**

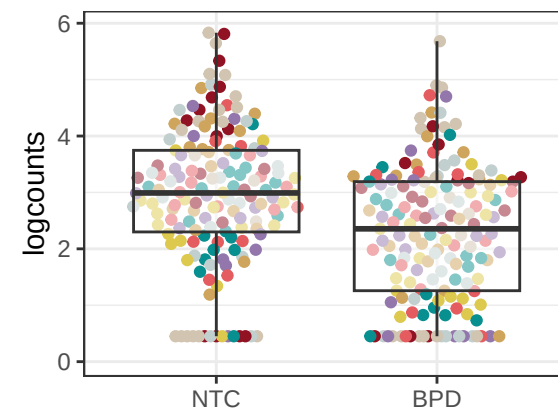
PRECAST (smoothed)

L-A sig: adj p < 2e-04 logFC = -1.01



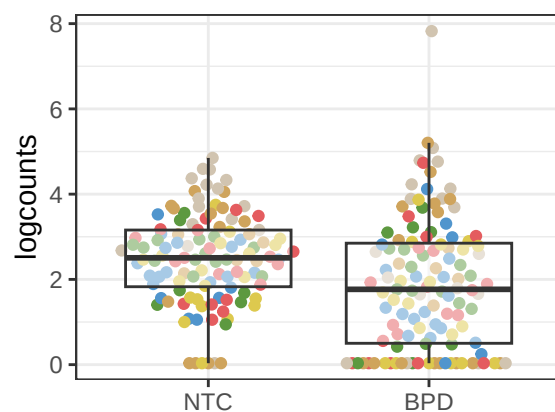
Seurat labels

L-A sig: adj p < 6e-04 logFC = -0.94

**TREM2**

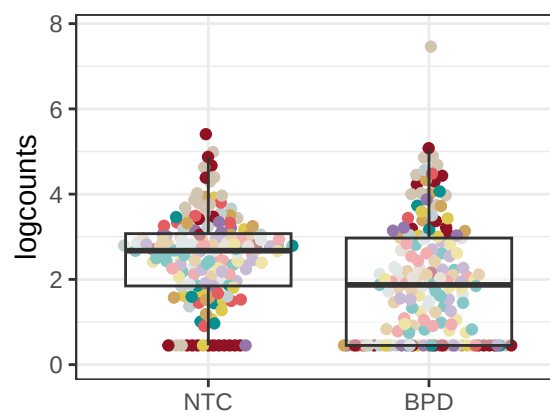
PRECAST (smoothed)

L-A sig: adj p < 3e-03 logFC = -1.01



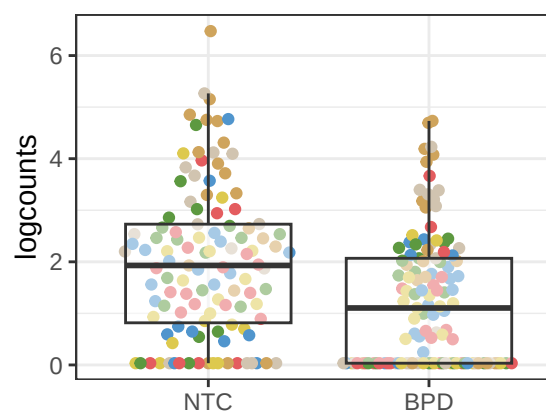
Seurat labels

L-A sig: adj p < 3e-03 logFC = -0.97

**P2RY12**

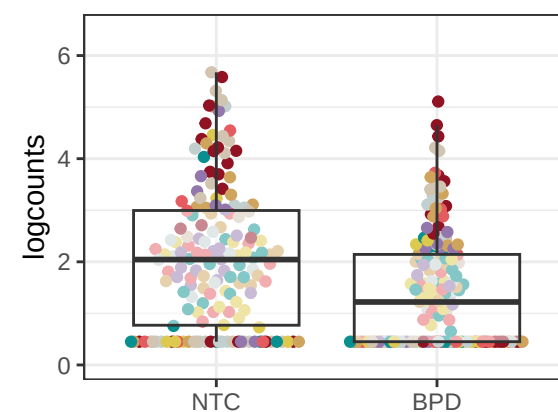
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.97



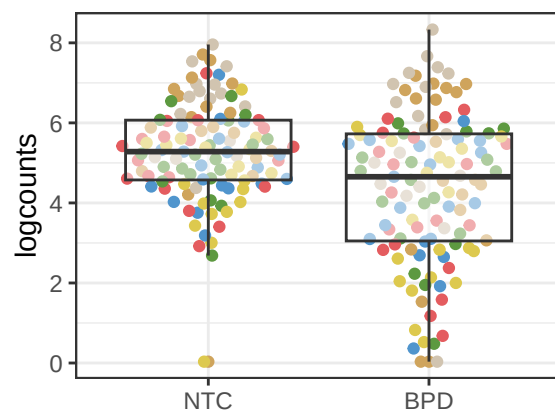
Seurat labels

L-A sig: adj p < 1e-02 logFC = -0.95

**C3**

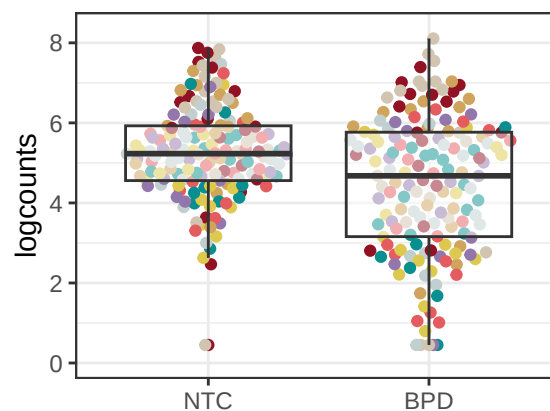
PRECAST (smoothed)

L-A sig: adj p < 3e-04 logFC = -0.97



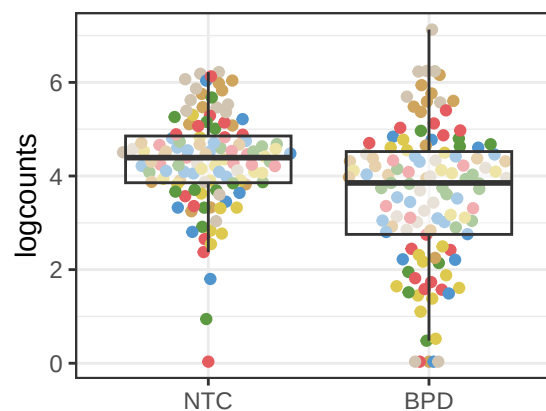
Seurat labels

L-A sig: adj p < 2e-05 logFC = -1

**CSF1R**

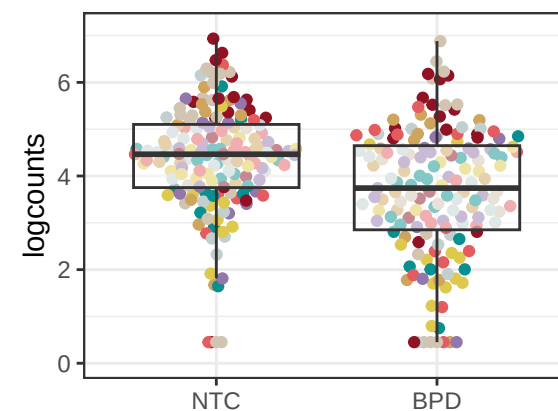
PRECAST (smoothed)

L-A sig: adj p < 1e-04 logFC = -0.93

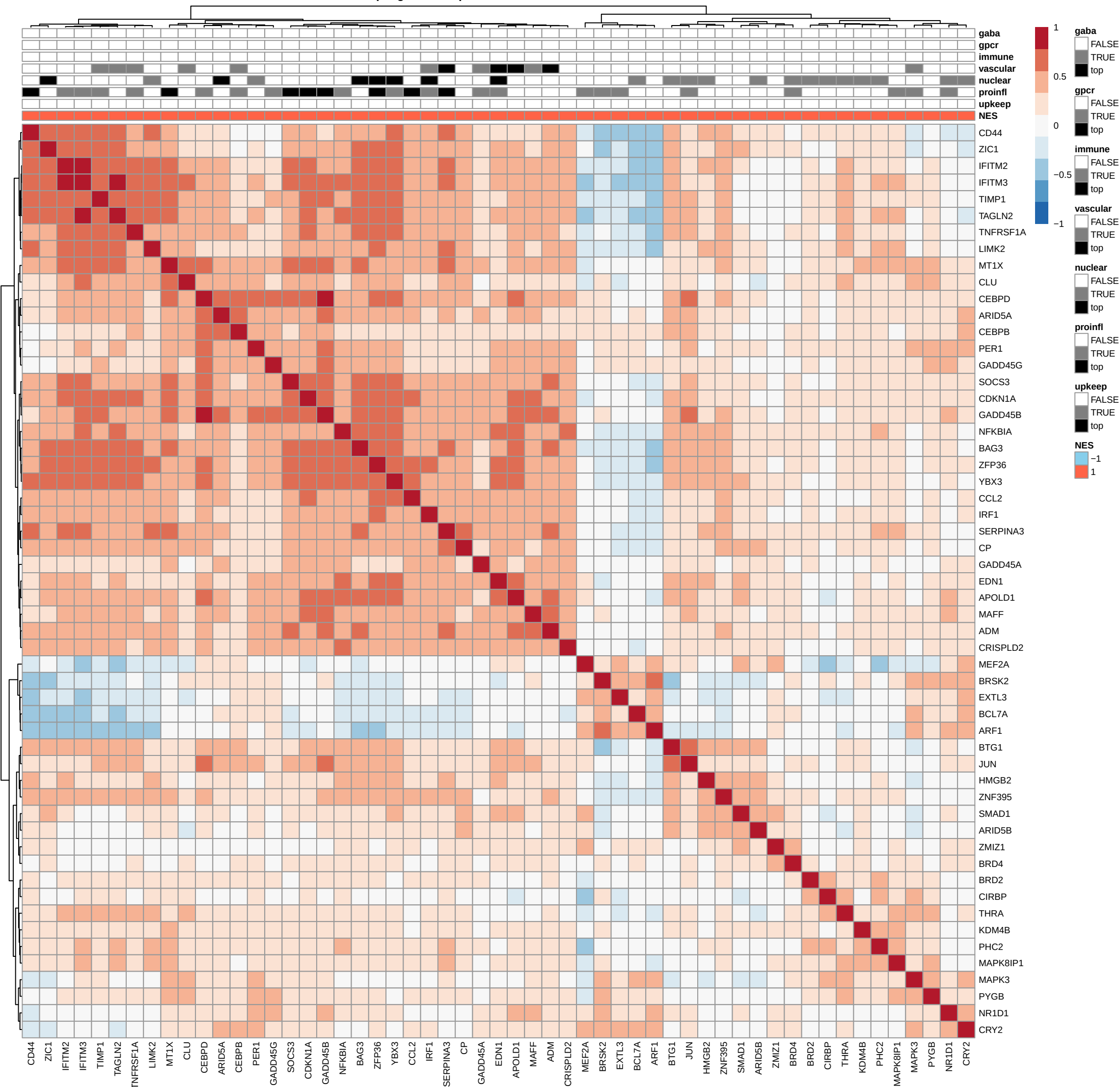


Seurat labels

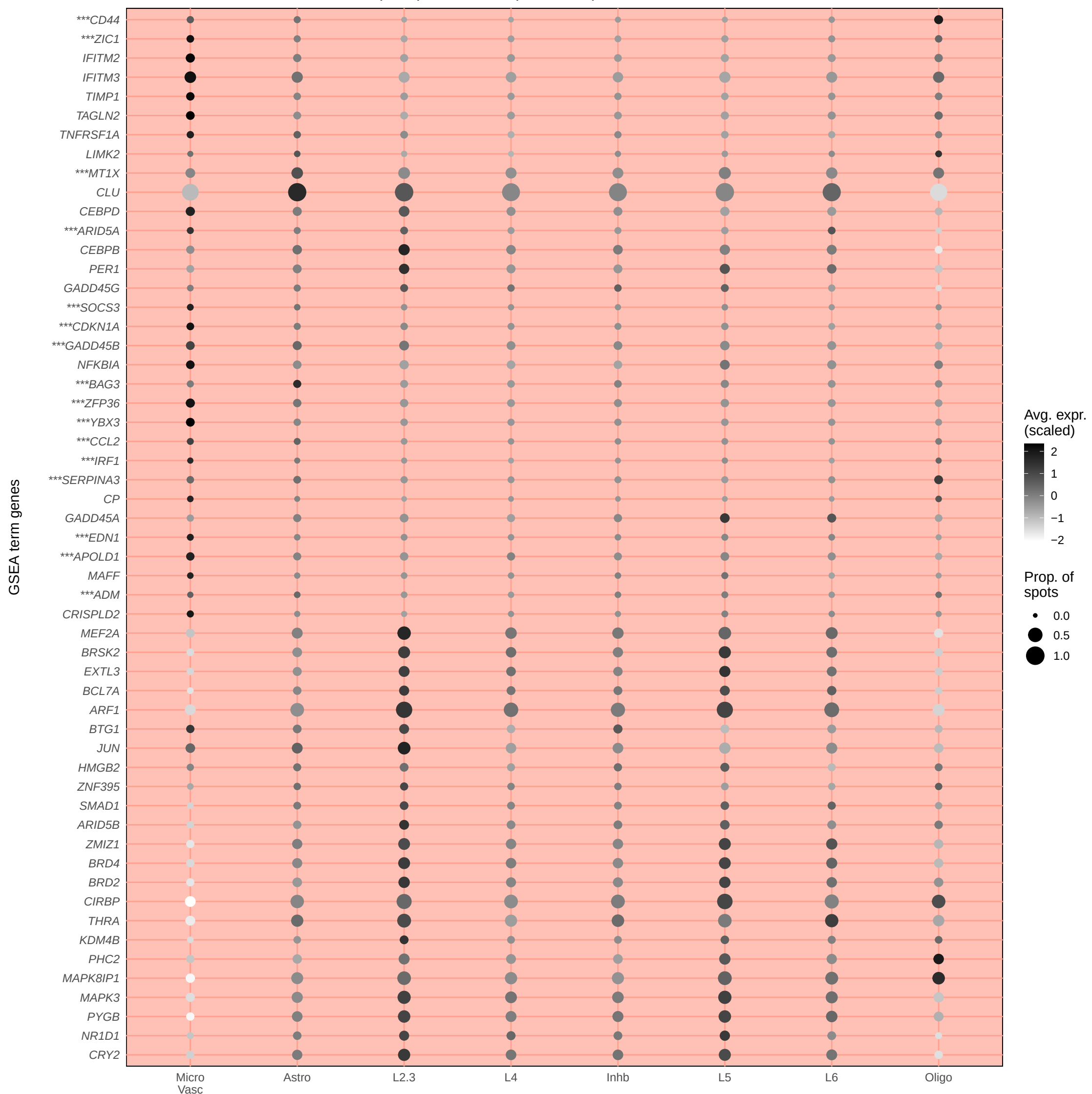
L-A sig: adj p < 3e-05 logFC = -0.9



NTC.BPD M L-A enriched genes
Top logFC DEGs per theme annotated with black

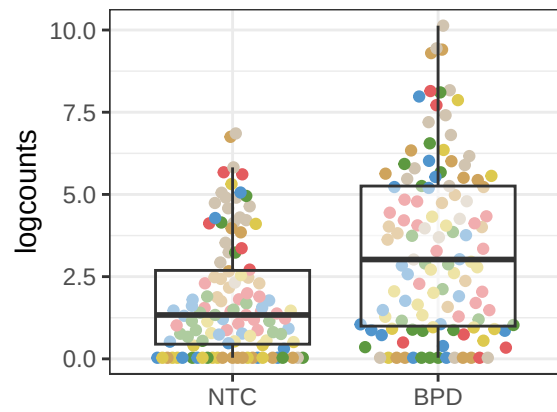


NTC.BPD M L-A enriched terms (***)top L-A DEGs per theme)

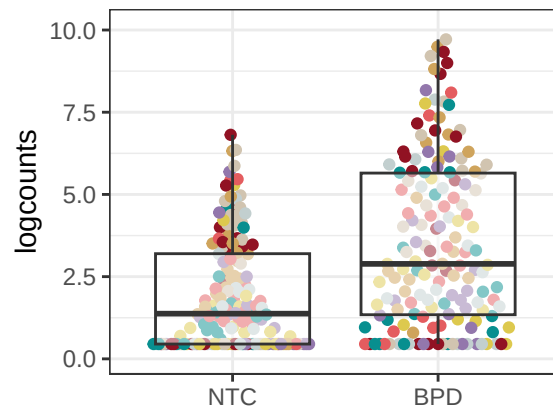


SERPINA3

PRECAST (smoothed)
L-A sig: adj p < 4e-03 logFC = 2.08

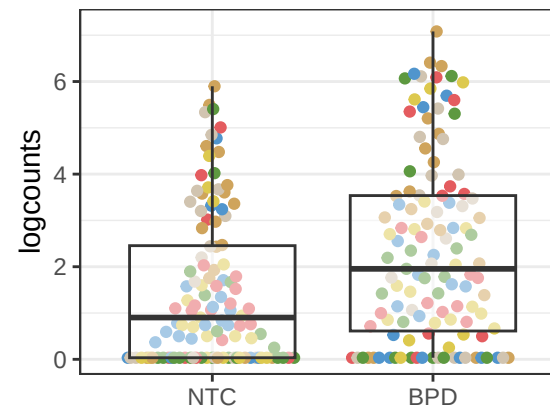


Seurat labels
L-A sig: adj p < 5e-05 logFC = 2.28

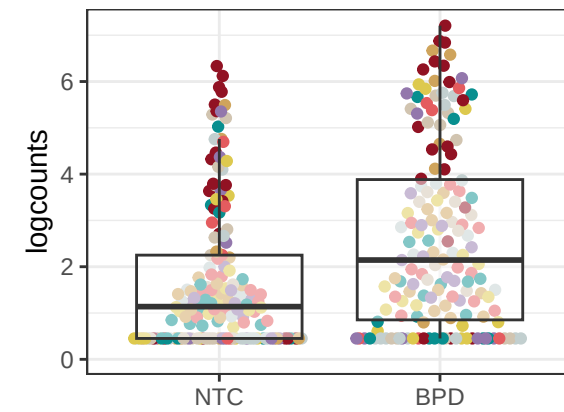


SOCS3

PRECAST (smoothed)
L-A sig: adj p < 1e-02 logFC = 1.44

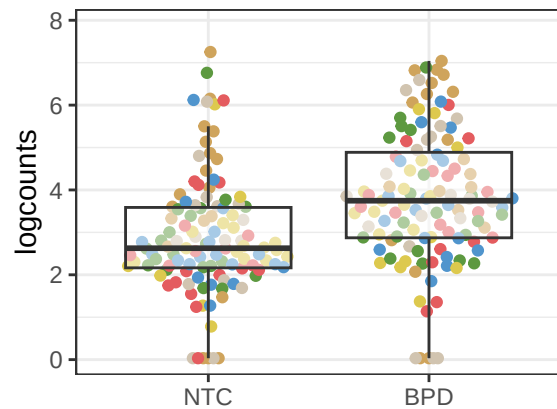


Seurat labels
L-A sig: adj p < 2e-03 logFC = 1.45

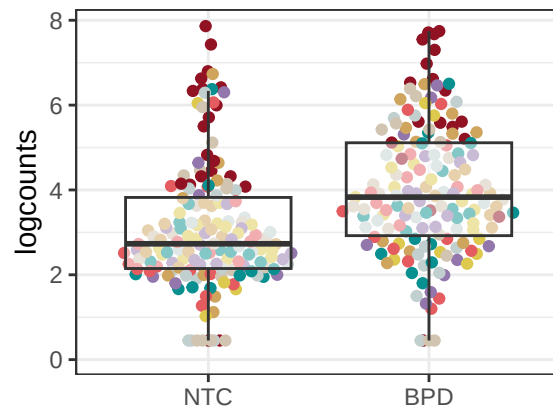


CDKN1A

PRECAST (smoothed)
L-A sig: adj p < 2e-04 logFC = 1.19

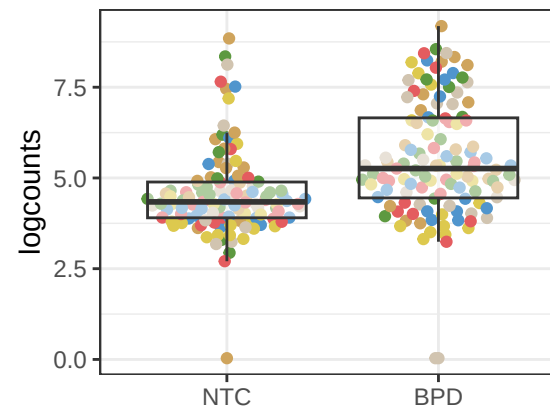


Seurat labels
L-A sig: adj p < 3e-06 logFC = 1.24

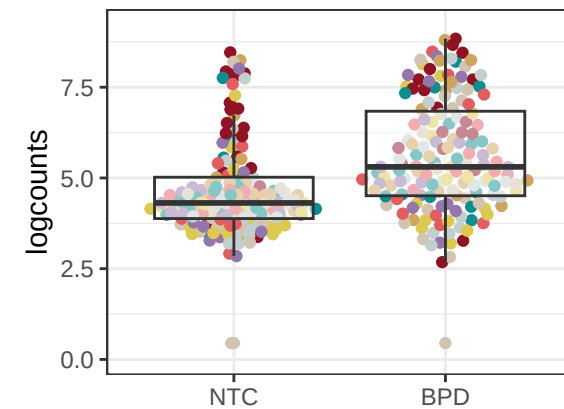


GADD45B

PRECAST (smoothed)
L-A sig: adj p < 2e-05 logFC = 1.16

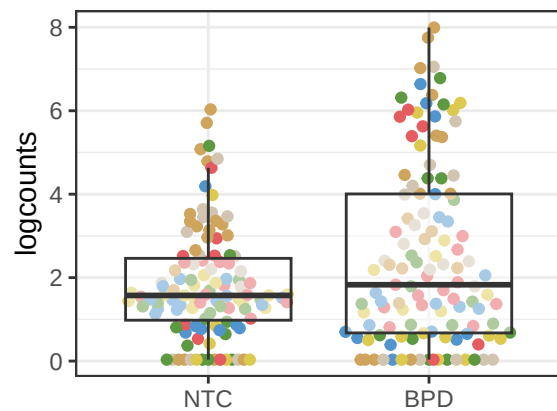


Seurat labels
L-A sig: adj p < 1e-07 logFC = 1.21

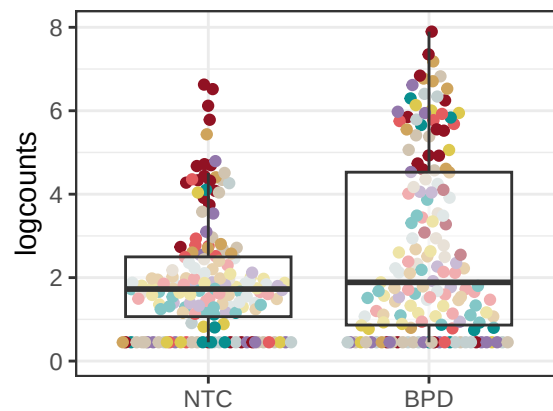


CCL2

PRECAST (smoothed)
L-A sig: adj p < 3e-02 logFC = 1.14

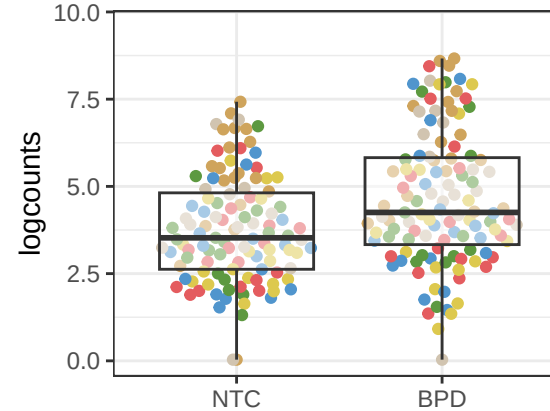


Seurat labels
L-A sig: adj p < 4e-03 logFC = 1.25

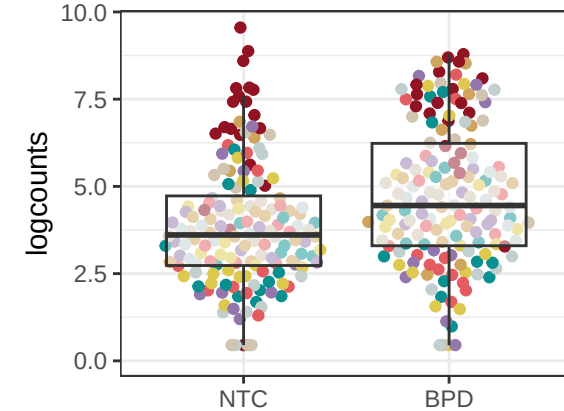


ZFP36

PRECAST (smoothed)
L-A sig: adj p < 1e-02 logFC = 1

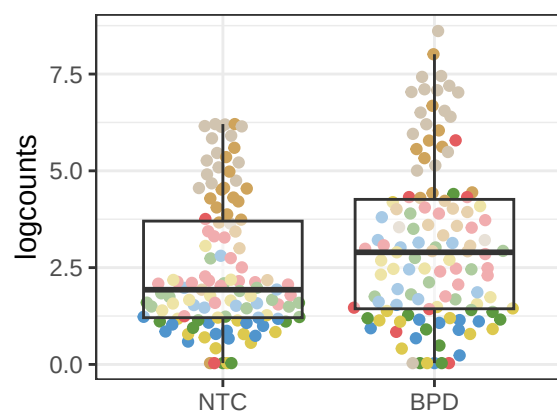


Seurat labels
L-A sig: adj p < 9e-03 logFC = 0.91

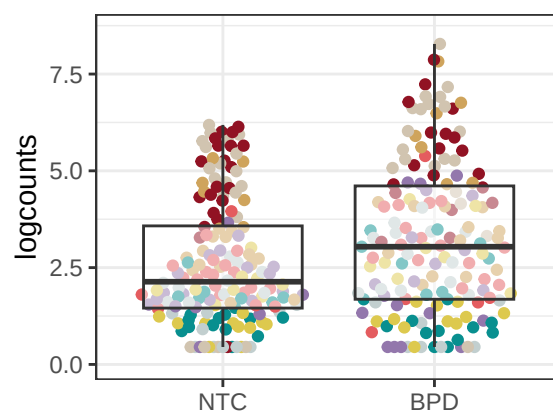


CD44

PRECAST (smoothed)
L-A sig: adj p < 3e-02 logFC = 0.88

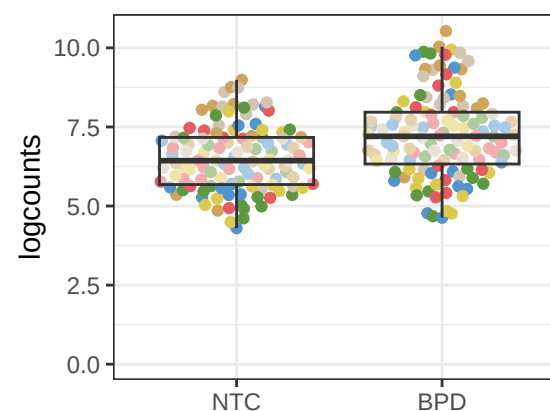


Seurat labels
L-A sig: adj p < 4e-03 logFC = 1.02

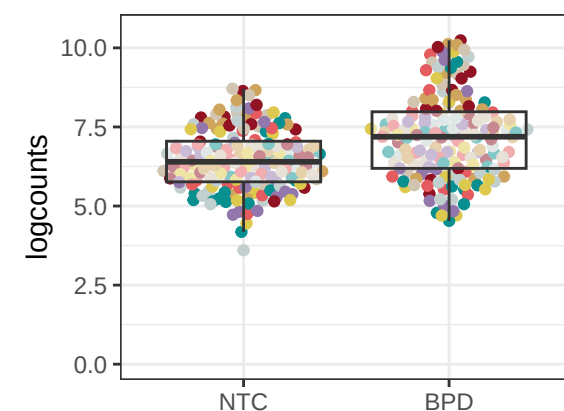


MT1X

PRECAST (smoothed)
L-A sig: adj p < 3e-04 logFC = 0.82



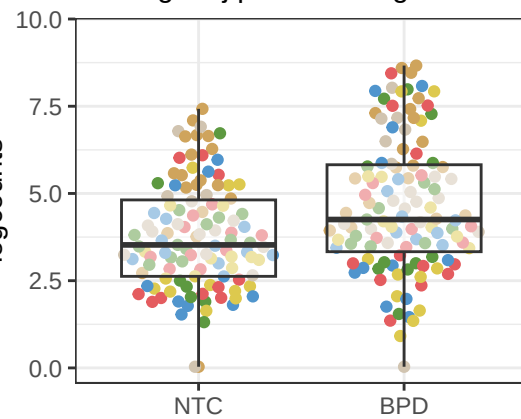
Seurat labels
L-A sig: adj p < 2e-05 logFC = 0.83



ZFP36

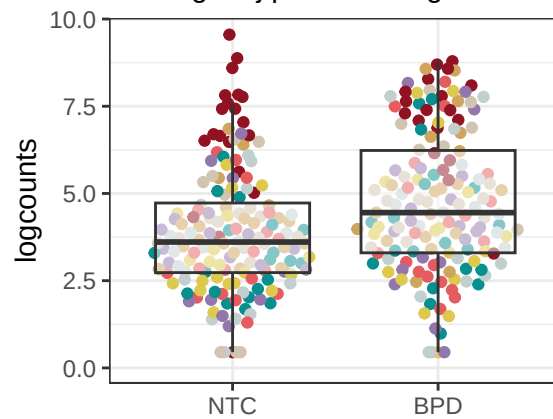
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = 1



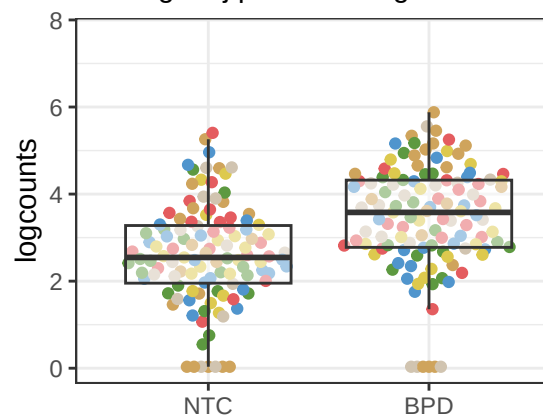
Seurat labels

L-A sig: adj p < 9e-03 logFC = 0.91

**EDN1**

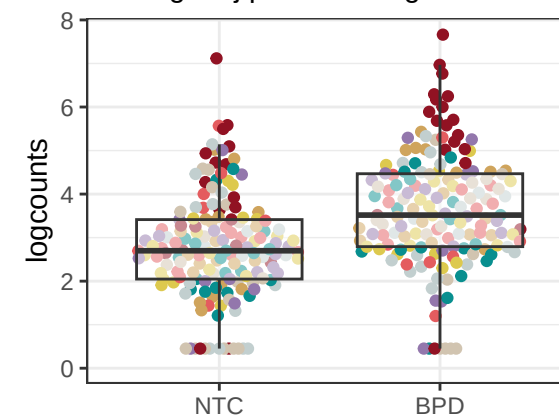
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = 0.81



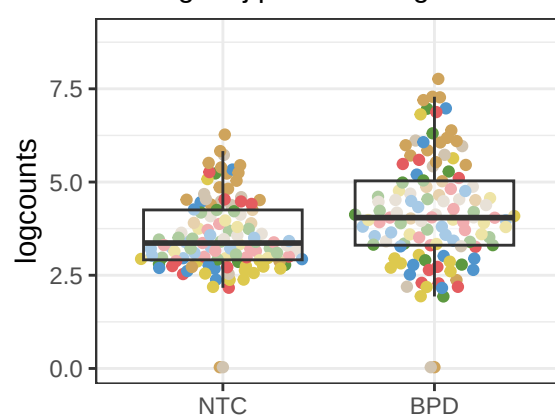
Seurat labels

L-A sig: adj p < 6e-04 logFC = 0.82

**YBX3**

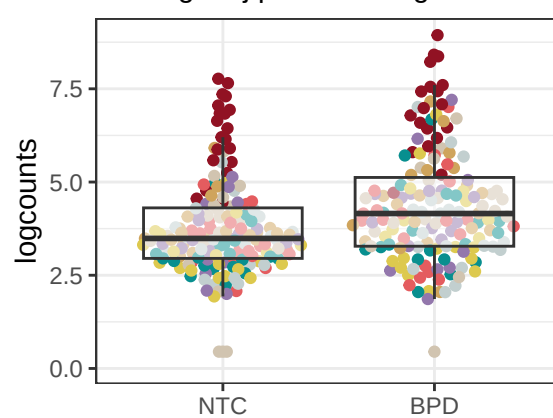
PRECAST (smoothed)

L-A sig: adj p < 1e-03 logFC = 0.81



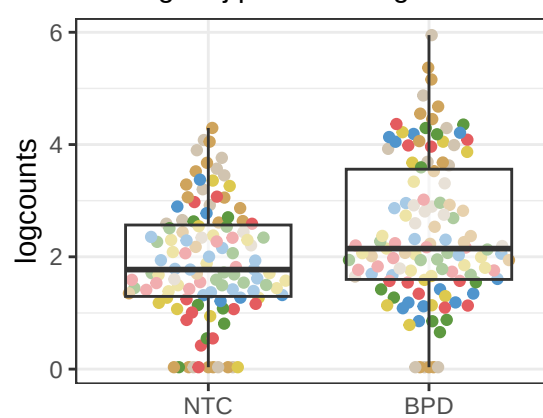
Seurat labels

L-A sig: adj p < 4e-04 logFC = 0.79

**IRF1**

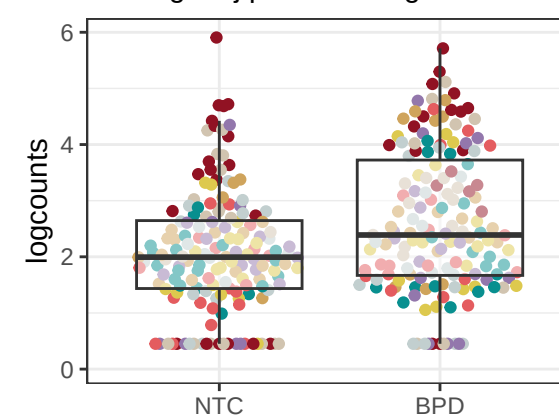
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = 0.72



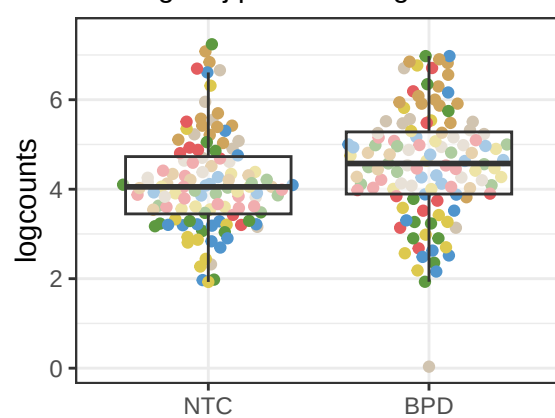
Seurat labels

L-A sig: adj p < 2e-02 logFC = 0.74

**BAG3**

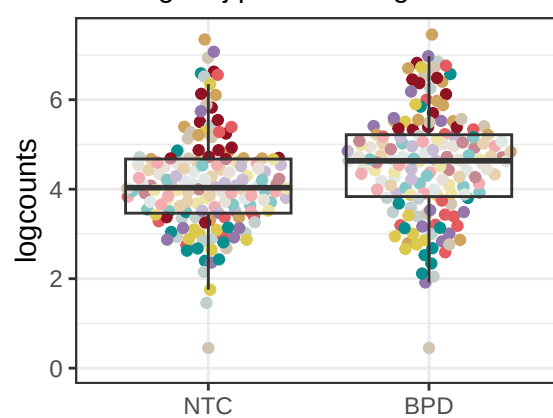
PRECAST (smoothed)

L-A sig: adj p < 4e-02 logFC = 0.6



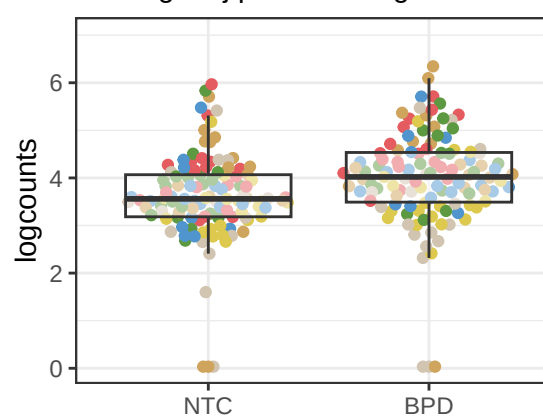
Seurat labels

L-A sig: adj p < 1e-02 logFC = 0.65

**ARID5A**

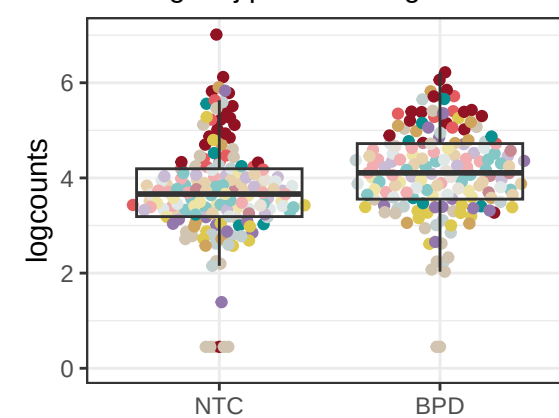
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = 0.55



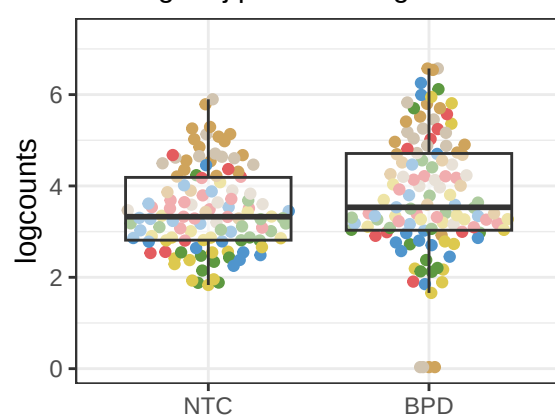
Seurat labels

L-A sig: adj p < 8e-04 logFC = 0.57

**ZIC1**

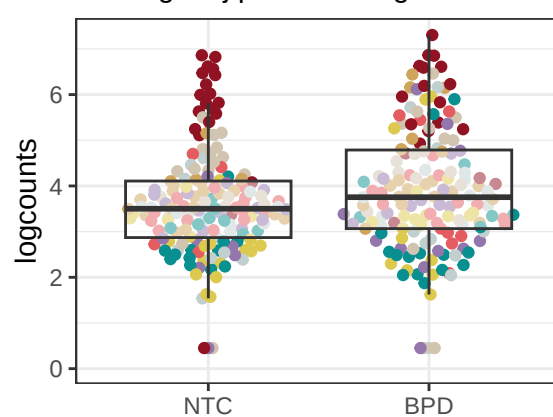
PRECAST (smoothed)

L-A sig: adj p < 4e-02 logFC = 0.51



Seurat labels

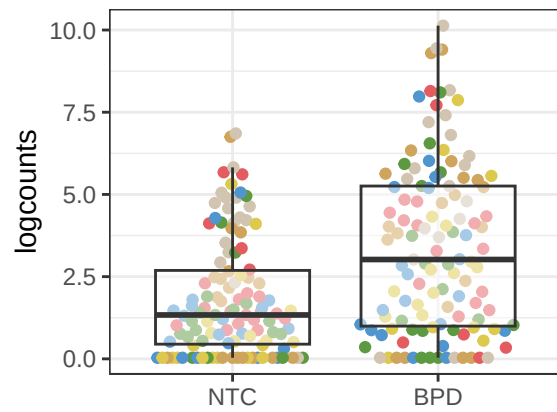
L-A sig: adj p < 3e-02 logFC = 0.53



SERPINA3

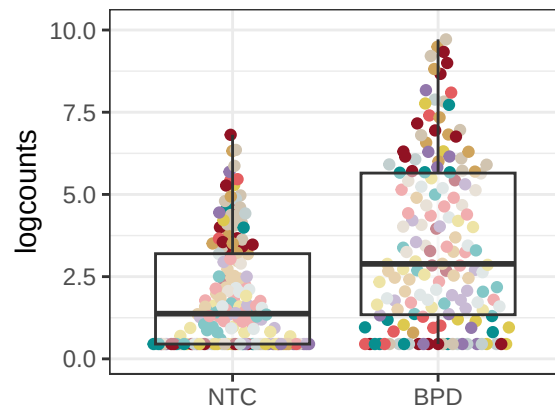
PRECAST (smoothed)

L-A sig: adj p < 4e-03 logFC = 2.08



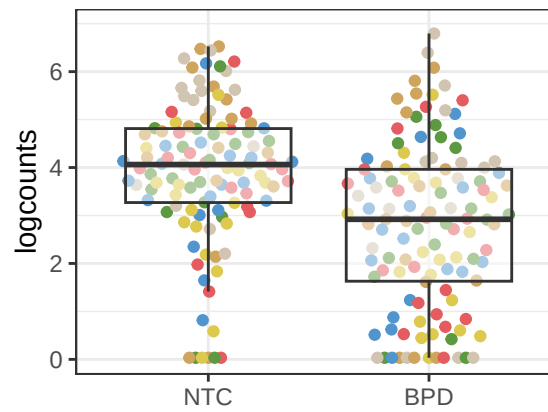
Seurat labels

L-A sig: adj p < 5e-05 logFC = 2.28

**CX3CR1**

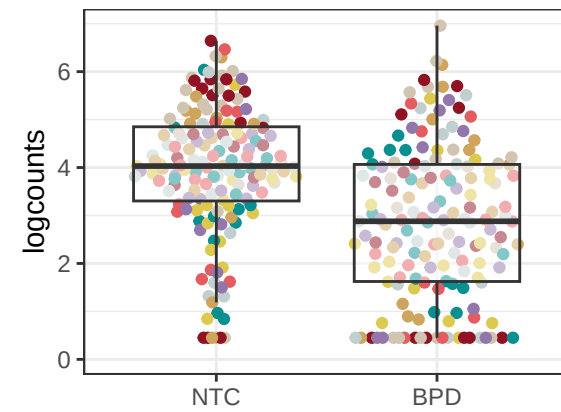
PRECAST (smoothed)

L-A sig: adj p < 1e-07 logFC = -1.52



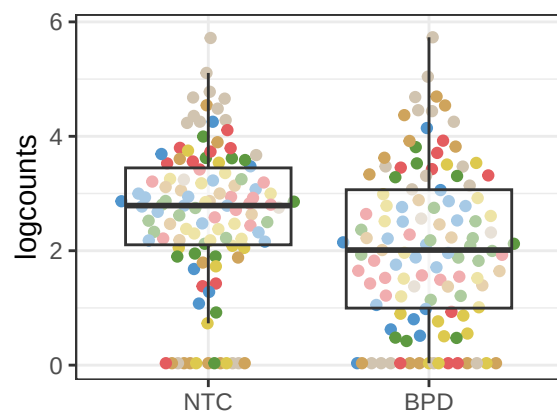
Seurat labels

L-A sig: adj p < 2e-08 logFC = -1.47

**SELPLG**

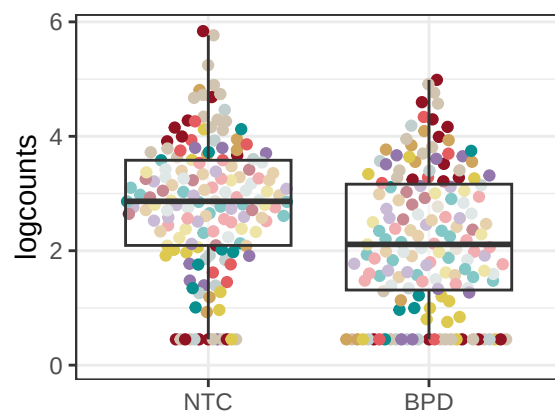
PRECAST (smoothed)

L-A sig: adj p < 9e-05 logFC = -1.22



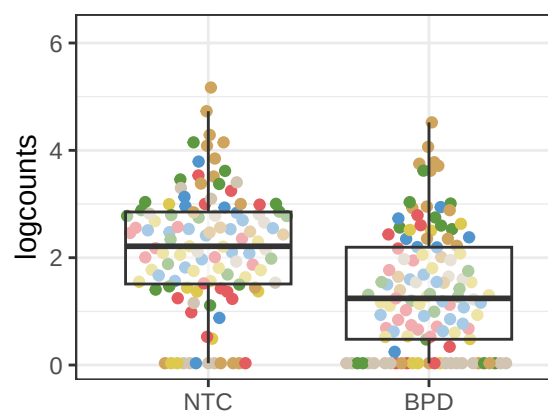
Seurat labels

L-A sig: adj p < 7e-05 logFC = -1.12

**FOLR2**

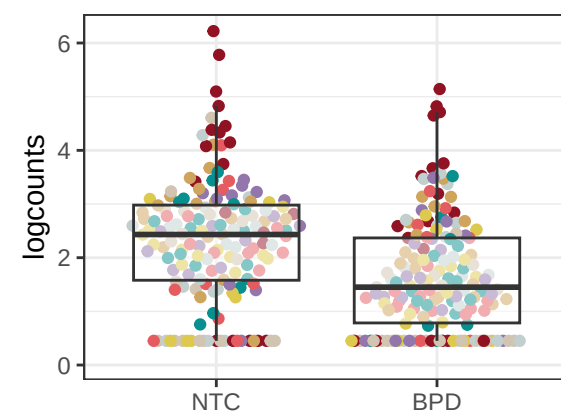
PRECAST (smoothed)

L-A sig: adj p < 2e-04 logFC = -1.1



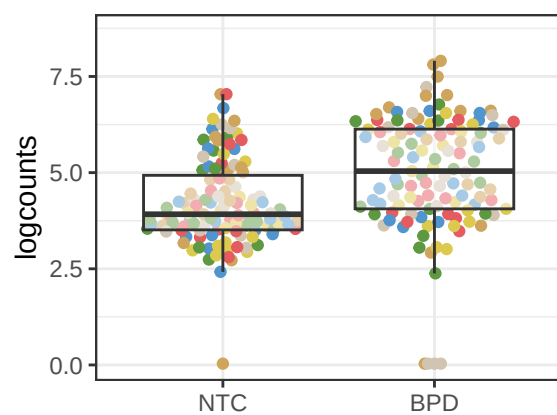
Seurat labels

L-A sig: adj p < 3e-04 logFC = -1.08

**APOLD1**

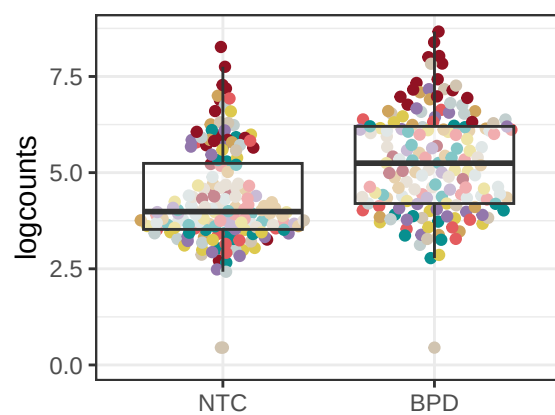
PRECAST (smoothed)

L-A sig: adj p < 2e-05 logFC = 0.95



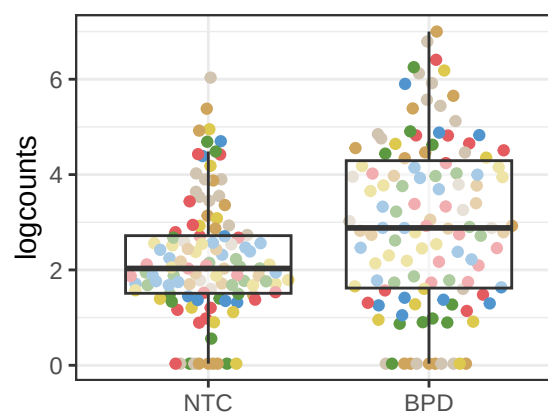
Seurat labels

L-A sig: adj p < 2e-06 logFC = 0.9

**ADM**

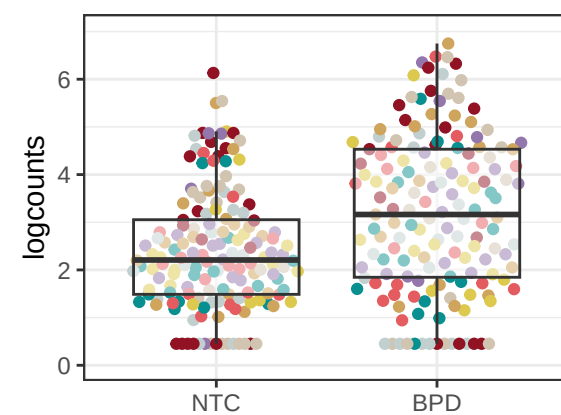
PRECAST (smoothed)

L-A sig: adj p < 3e-02 logFC = 0.94



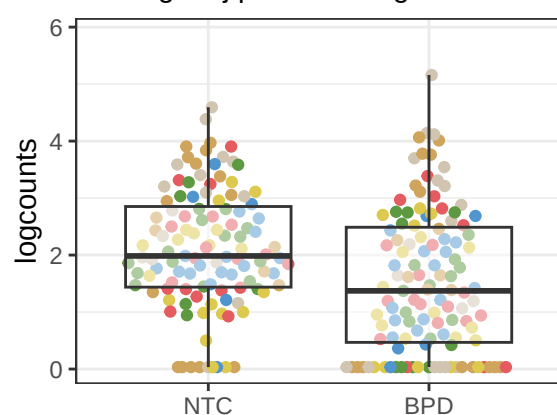
Seurat labels

L-A sig: adj p < 4e-03 logFC = 1.05

**C3AR1**

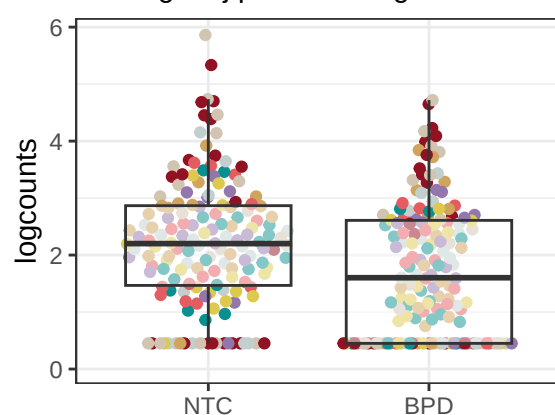
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.87



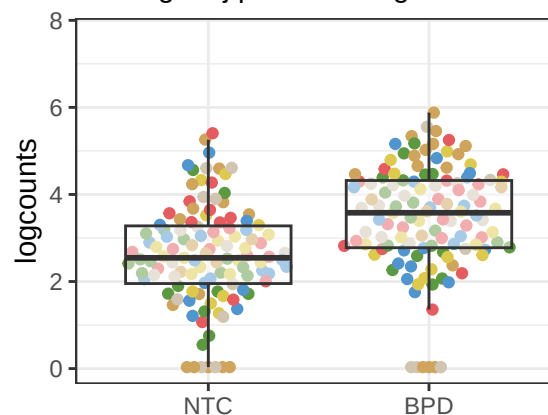
Seurat labels

L-A sig: adj p < 3e-03 logFC = -1

**EDN1**

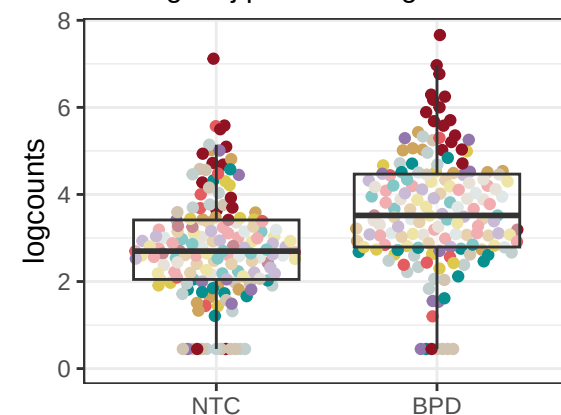
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = 0.81

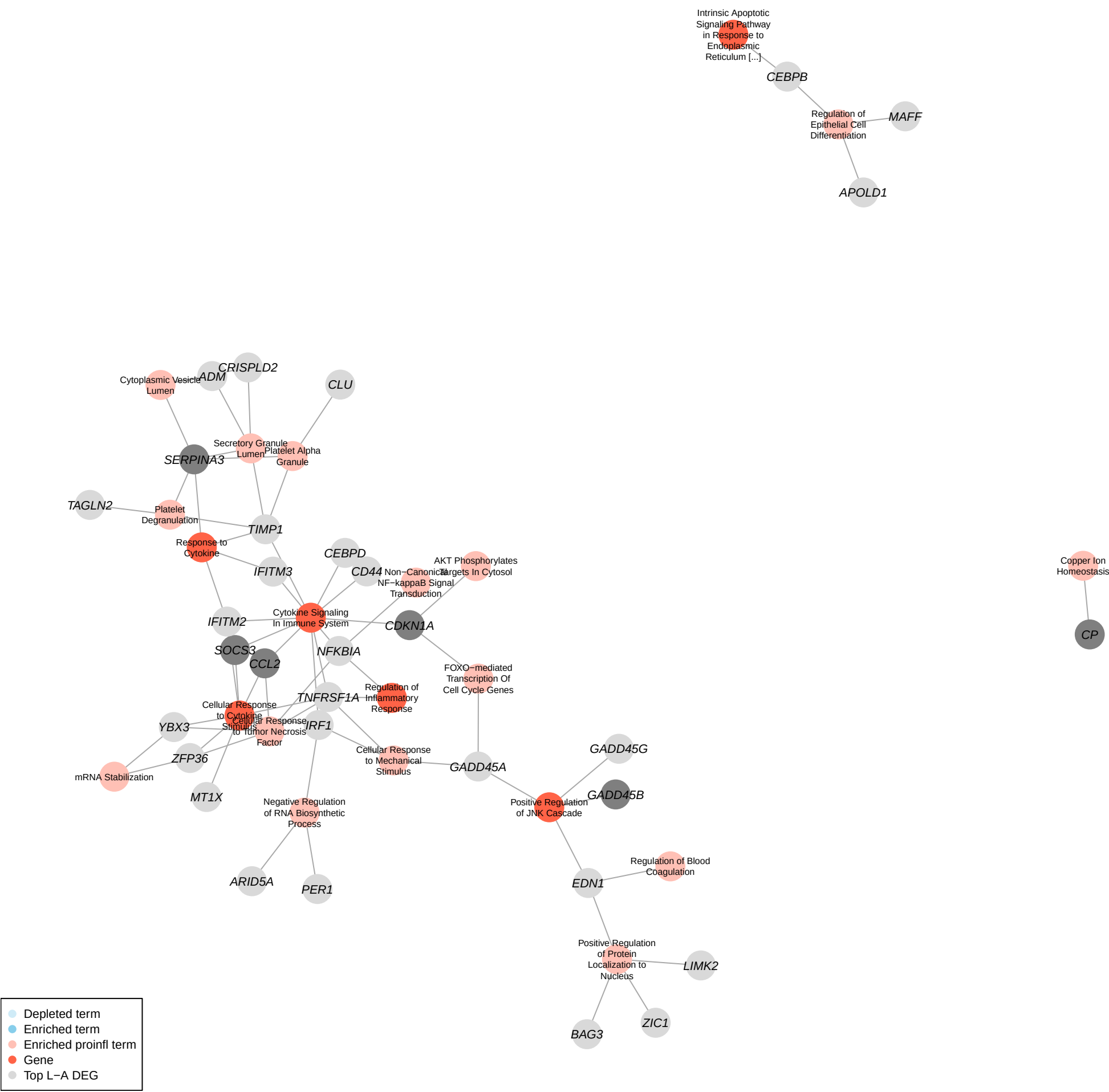


Seurat labels

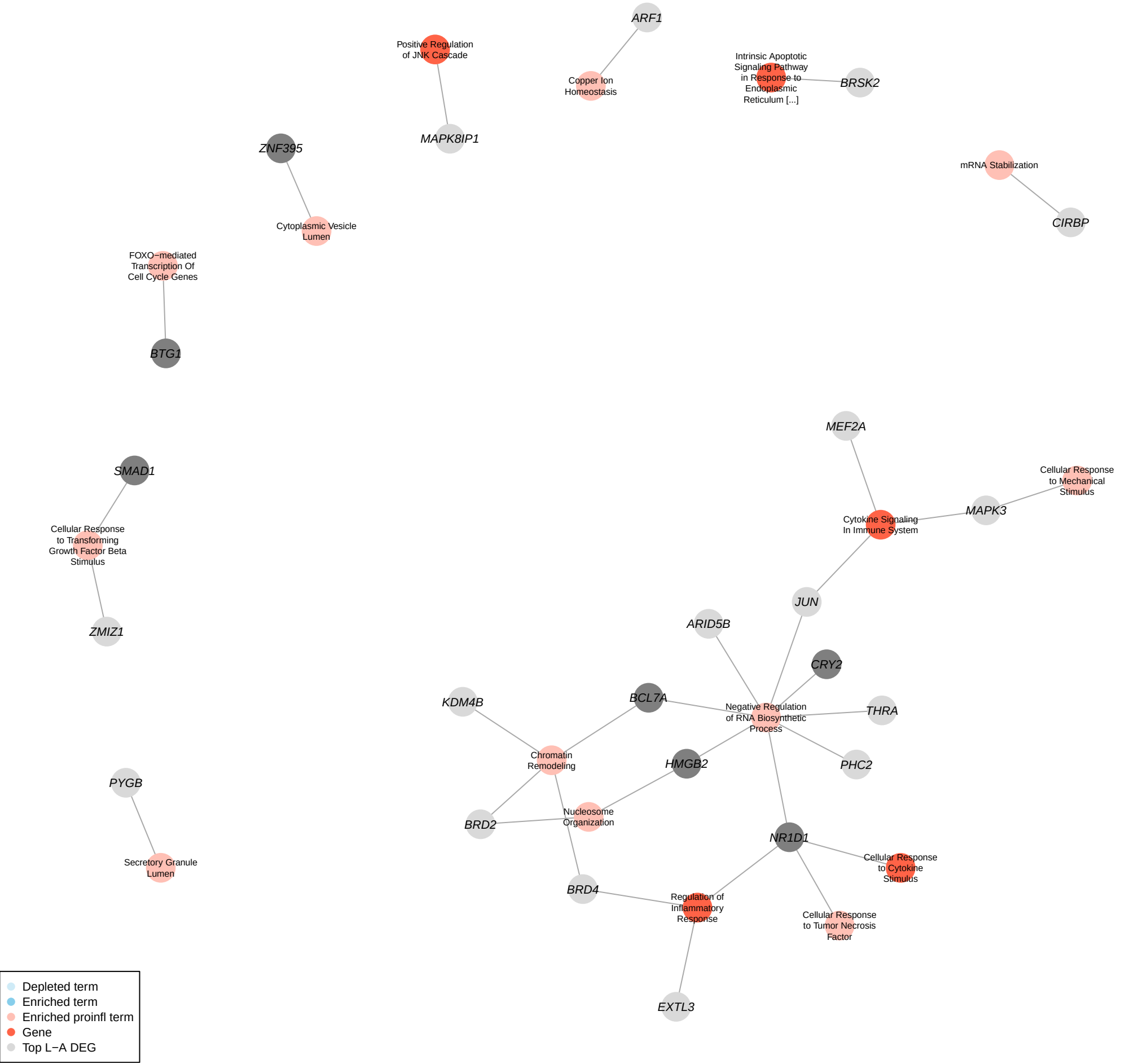
L-A sig: adj p < 6e-04 logFC = 0.82



NTC.BPD M enriched (glial)

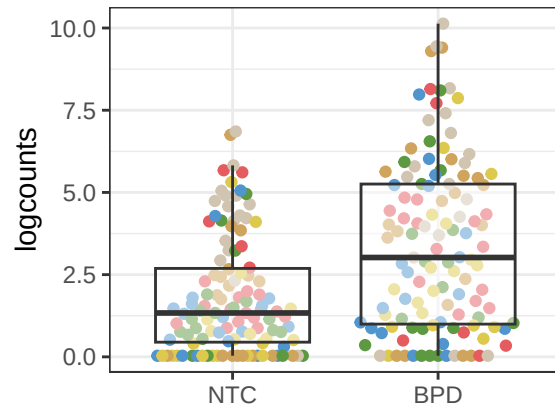


NTC.BPD M enriched (neuronal)

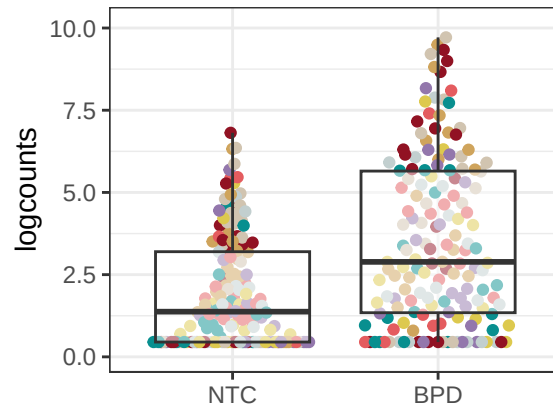


SERPINA3

PRECAST (smoothed)
L-A sig: adj p< 4e-03 logFC = 2.08

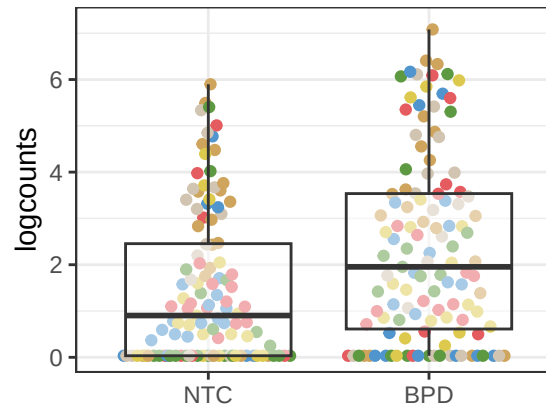


Seurat labels
L-A sig: adj p< 5e-05 logFC = 2.28

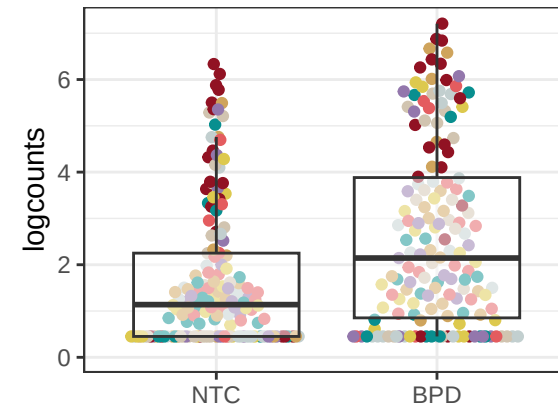


SOCS3

PRECAST (smoothed)
L-A sig: adj p< 1e-02 logFC = 1.44

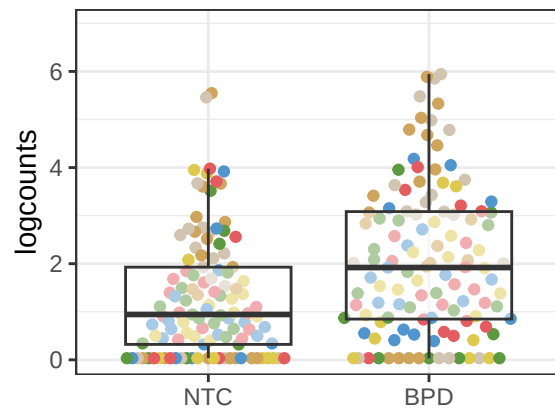


Seurat labels
L-A sig: adj p< 2e-03 logFC = 1.45

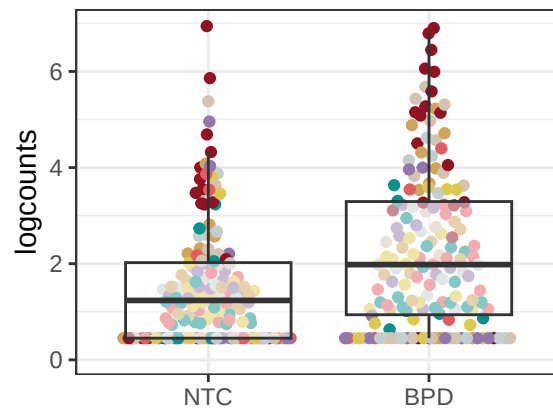


CP

PRECAST (smoothed)
L-A sig: adj p< 2e-02 logFC = 1.23

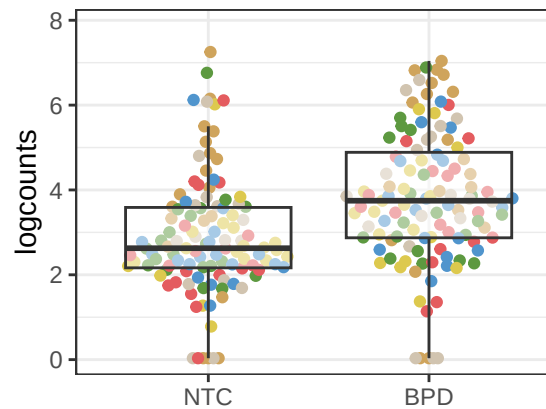


Seurat labels
L-A sig: adj p< 4e-03 logFC = 1.27

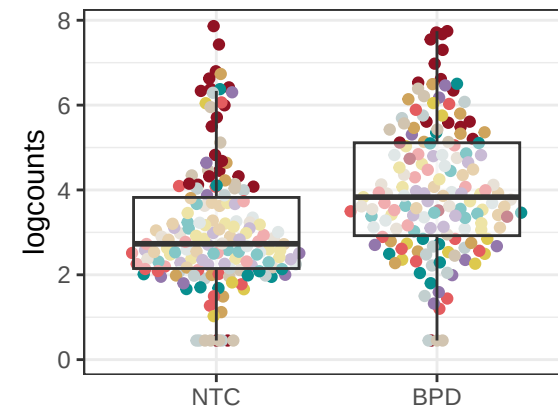


CDKN1A

PRECAST (smoothed)
L-A sig: adj p< 2e-04 logFC = 1.19

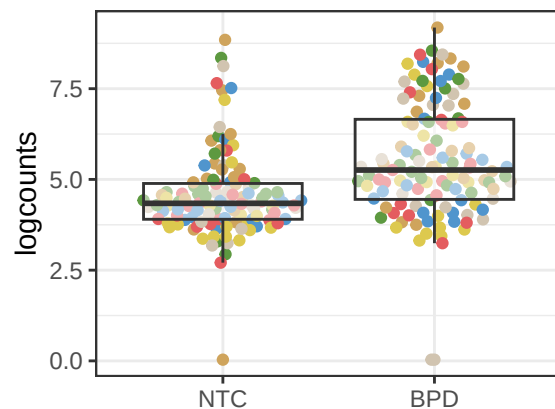


Seurat labels
L-A sig: adj p< 3e-06 logFC = 1.24

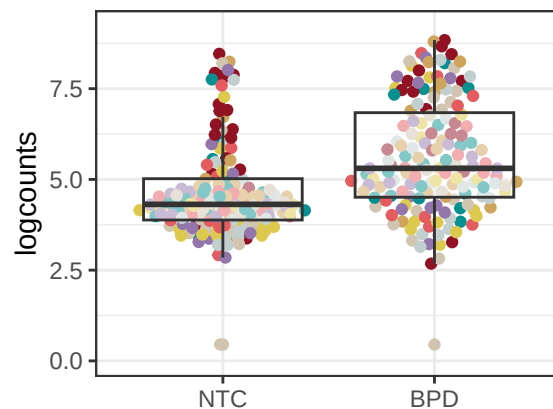


GADD45B

PRECAST (smoothed)
L-A sig: adj p< 2e-05 logFC = 1.16

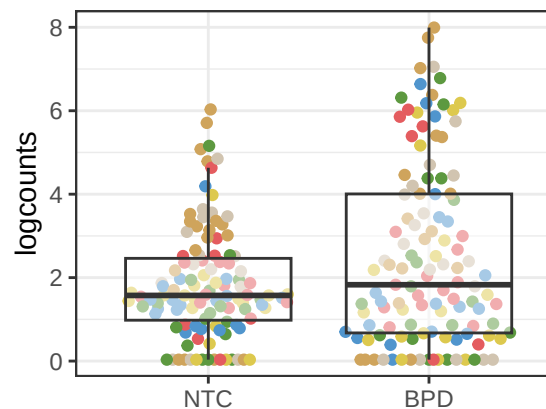


Seurat labels
L-A sig: adj p< 1e-07 logFC = 1.21

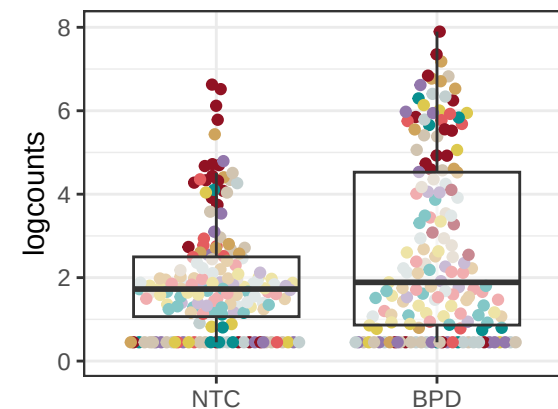


CCL2

PRECAST (smoothed)
L-A sig: adj p< 3e-02 logFC = 1.14



Seurat labels
L-A sig: adj p< 4e-03 logFC = 1.25

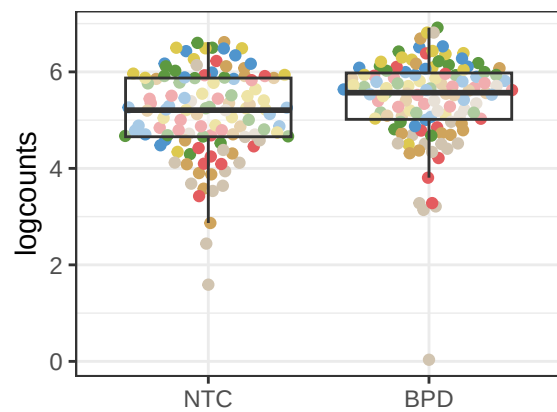


Neuronal enriched top genes

NR1D1

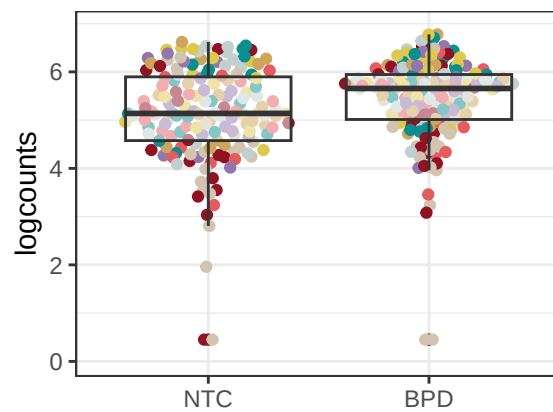
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = 0.39



Seurat labels

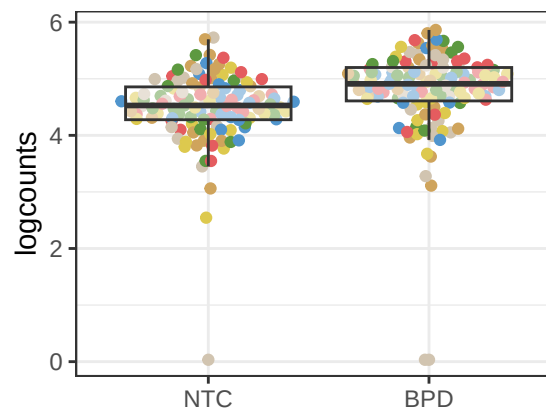
L-A sig: adj p < 1e-02 logFC = 0.38



SMAD1

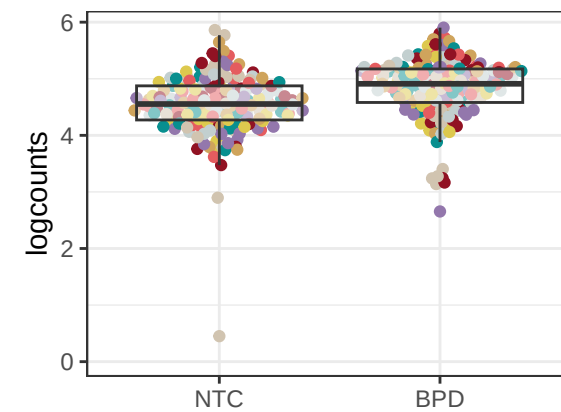
PRECAST (smoothed)

L-A sig: adj p < 7e-04 logFC = 0.36



Seurat labels

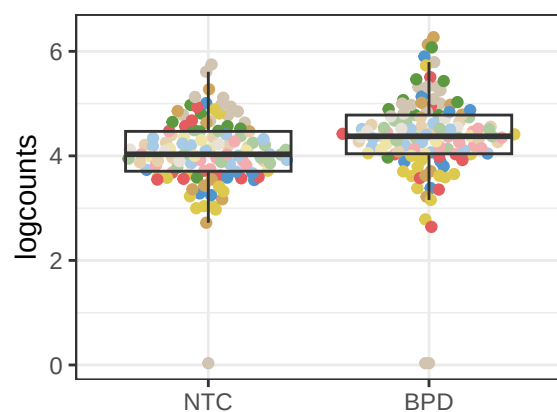
L-A sig: adj p < 7e-04 logFC = 0.37



ZNF395

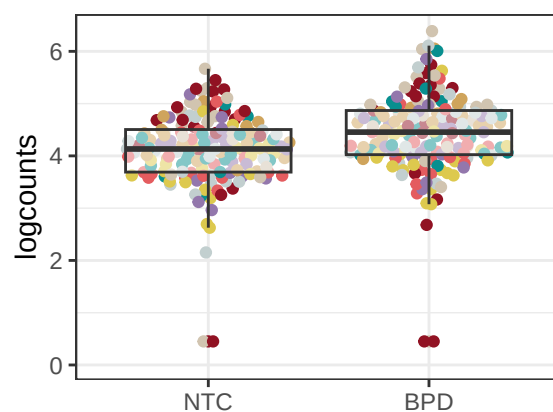
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = 0.35



Seurat labels

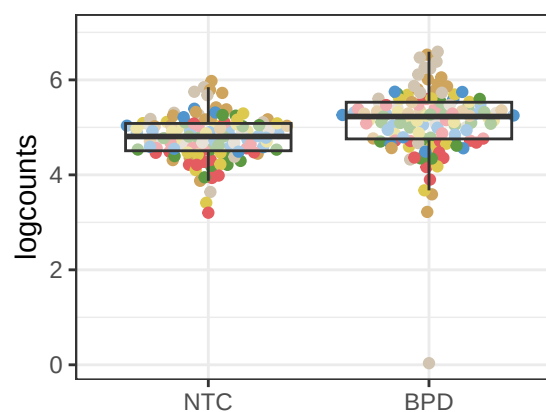
L-A sig: adj p < 4e-02 logFC = 0.36



HMGB2

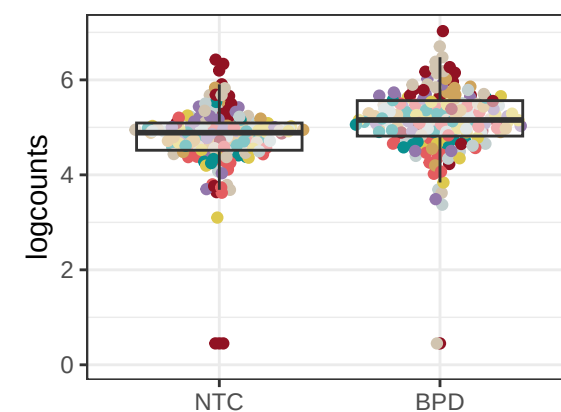
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = 0.3



Seurat labels

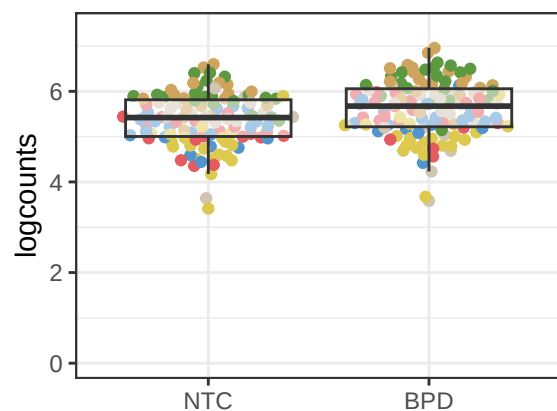
L-A sig: adj p < 2e-02 logFC = 0.31



BTG1

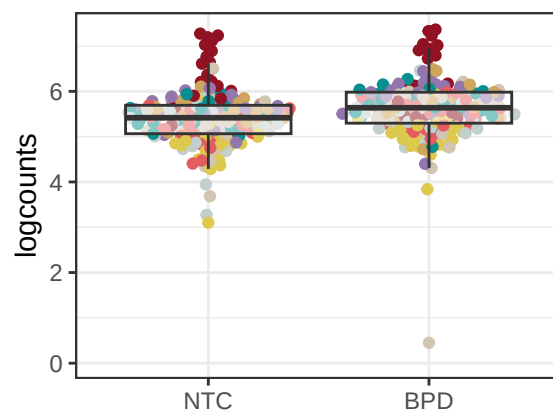
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = 0.26



Seurat labels

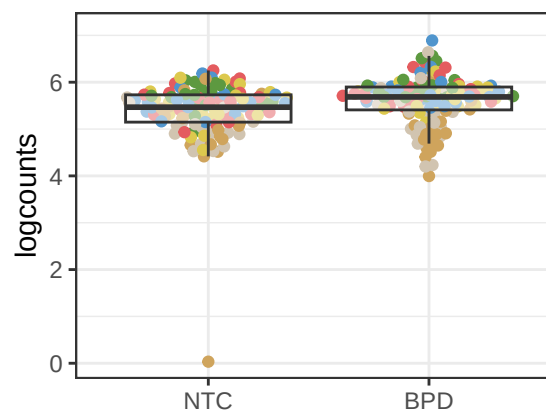
L-A sig: adj p < 1e-02 logFC = 0.28



BCL7A

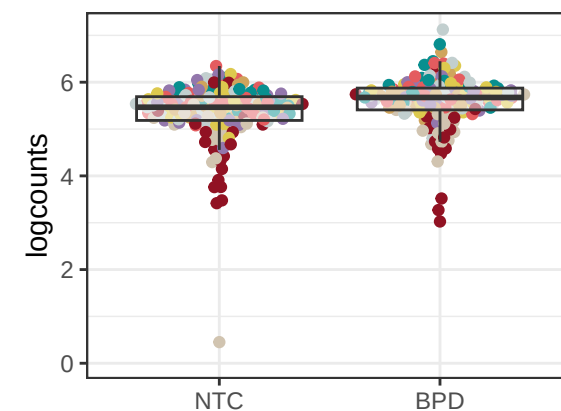
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = 0.26



Seurat labels

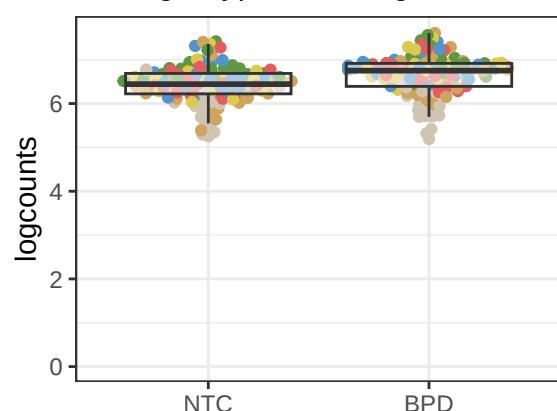
L-A sig: adj p < 4e-02 logFC = 0.24



CRY2

PRECAST (smoothed)

L-A sig: adj p < 6e-03 logFC = 0.23



Seurat labels

L-A sig: adj p < 2e-03 logFC = 0.25

